

Appendix of "Global Uncertainty" (Caggiano and Castelnovo 2022)

This Appendix collects further information with respect to the one in our "Global Uncertainty" paper. In particular:

- Section A deals with the list of countries in our dataset, the geographical coverage, and offers details on the computation of the financial volatilities we deal with.
- Section B offers a first look at the volatility data.
- Section C shows that the Dynamic Hierarchical Factor Model (DHFM) we work with to estimate the Global Uncertainty Factor (GFU) can be interpreted as a time-varying intercept-model.
- Section D offers details on our algorithm to estimate the DHFM framework with an unbalanced panel.
- Section E contrasts our estimated GFU factor with the one constructed as a weighted average of the volatility measures in our sample as well as the one estimated by principal components. It also uses two alternative approaches to measuring volatility: realized bipower variation and log-volatilities.
- Section F comments on our regional and country-specific factors.
- Section G plots the data used in our VAR analysis, the median realization of the GFU shocks and the identified dates for the narrative restrictions, and the VAR impulse responses.
- Section H contrasts our estimate of the GFU obtained with the whole sample of available data (1992M7-2020M5) with the estimate we obtained by dropping the COVID-19 observations and ending the sample in 2019M6.
- Section I analyzes the role of different seeds for our stochastic simulations (in first place, for the algorithm we use to draw orthonormal rotation matrices to explore the set of admissible models conditional on the restriction we impose to identify macroeconomic shocks in our VAR analysis).

- Section J investigates the role played by the rotation matrices *per se* in influencing our results.
- Section K shows that our baseline results are robust to doubling the number of draws (candidate models) and to weakening the Great Recession-related requirement for a GFU shock to be identified as such.
- Section L documents the IRFs obtained by replacing Miranda-Agrippino and Rey’s (2020) global financial cycle indicator in our VAR analysis with a novel measure of global financial cycle estimated with our dynamic factor model.
- Section M documents the robustness of our IRFs to changing the number of lags of the VAR.
- Section N proposes a placebo exercise to show that the component of financial uncertainty our model assigns to Latin America that is orthogonal to GFU exerts effects neither on the global financial cycle nor on world output.
- Section O documents the significance of the correlation of our GFU shocks with respect to two different proxies for US uncertainty, a proxy for financial uncertainty shocks and a proxy for macroeconomic uncertainty shocks.
- Section P provides details on the reduced-form - structural VAR mapping.
- Section Q compares moments implied by our non-recursive VAR with those implied by a recursive one.
- Section R offers a brief description on the construction of the WIP index by Baumeister and Hamilton (2019).
- Section S documents that our results are robust to dropping the ratio restriction that requires a first-moment financial shock to have a quantitatively larger on-impact effect on the global financial cycle than on global uncertainty.

A: Our dataset: Geographical coverage and computation of financial volatilities

As stated in the paper, the dataset comprises 42 countries that account for about 83% of world GDP in 2019 (measured in PPP) and cover five regions (North America, Latin

America, Europe, Asia, and Oceania). Table A1 reports the list of countries, and Figure A1 depicts the corresponding geographical coverage.

We move from daily returns to monthly volatilities by computing the latter as follows. Let x_{rcntd} be the raw observation of region r , country c , variable n , month t , and day d . Define daily stock market and exchange rate returns as $ret_{rcntd} \equiv \ln x_{rcntd} - \ln x_{rcnt(d-1)}$, and government bond yields as $ret_{rcntd} \equiv x_{rcntd}$. Then, the monthly realized volatility of variable n for country c belonging to region r at month t is given by

$$\tilde{Z}_{rcnt} = \sqrt{\frac{1}{M_{rcnt} - 1} \sum_{d=1}^{M_{rcnt}} (ret_{rcntd} - \overline{ret}_{rcnt})^2} \quad (1)$$

where M_{rcnt} indicates the number of observations of variable n for country c belonging to region r available in month t , and the month-specific mean $\overline{ret}_{rcnt} = \frac{1}{M_{rcnt}} \sum_{d=1}^{M_{rcnt}} ret_{rcntd}$. The month-specific mean is handy to control for the large swings in financial returns that occurred during extreme events such as the global financial crisis or the COVID-19 pandemic. Monthly volatilities $\tilde{Z}_{rcnt}$ are then standardized to obtain the zero mean, unit standard deviation observables Z_{rcnt} , which are used to estimate our factor model. While the three financial series mentioned above are available for most countries, our dataset does not comprise 10-year Government bond yields (again, at a daily frequency) for Argentina and three Asian countries (Honk Kong, Pakistan, and the Philippines). Moreover, it should be noted that not for all available series we have all observations spanning the July 1992 - May 2020 sample. As pointed out in the paper, our unbalanced panel features 122 series for a total of 37,710 observations

B: A first look at the monthly volatility data

To appreciate cross-country similarities and differences in financial volatility, Figure A2 (upper panel) plots the volatility of stock market returns for selected countries from 1992 to 2020. In correspondence of the GFC and the explosion of the COVID-19 pandemic, all volatilities series spike upward simultaneously. Their average pairwise correlation is 0.44. The evidence in the Figure points to the possibility of a global component in financial volatility. At the same time, when looking at these data, one can appreciate the importance of modelling regional and country-specific dynamics. Figure A2 (bottom panels) displays selected pairwise correlations. Stock market volatilities in the US and Canada - countries belonging to the same region, i.e., North America - are highly correlated, possibly because of the role of the US economy as "hegemon" in the region

(and, more broadly, internationally). Despite being significantly positive, the correlation between the US and Greece is weaker than the US-Canada one, indicating that for Greece regional or country-specific factors may also be at play. This becomes even more evident when looking at the low correlation the US and China stock market volatilities, which clearly points to the existence of important country-specific components. We take this evidence as supportive of a multi-level factor approach featuring global, regional, and country-specific components.

C: Coefficients restrictions and time-varying intercepts in the DHFM

Consider the four level factor model of Section 2.2:

$$\begin{aligned} Z_{rcnt} &= \lambda_{C,rc}^n(L) C_{rct} + e_{Z_{rcnt}} \\ C_{rct} &= \Lambda_{R.rc}(L) R_{rt} + e_{C_{rct}} \\ R_{rt} &= \Lambda_{GFU,r}(L) GFU_t + e_{R_{rt}} \\ \Psi_{GFU}(L) GFU_t &= e_{GFU_t} \end{aligned}$$

where GFU_t is the common factor, $\Lambda_{GFU,r}(L)$ is the distributed lag of loadings on the common factor, $e_{R_{rt}}$ is the region-specific variation, R_{rt} is the $(k_R \times 1)$ vector of region-specific factors, where k_R denotes the number of regions, $\Lambda_{R.rc}(L)$ is the distributed lag of loadings on the region-specific factors, $e_{C_{rct}}$ is the country-specific variation, C_{rct} is the $(K_{Cr} \times 1)$ vector of country-specific factors, where K_{Cr} denotes the number of countries in region r , $\lambda_{C,rc}^n(L)$ is the distributed lag of loadings on the country-specific factors, and $e_{Z_{rcnt}}$ is the series specific variation. The idiosyncratic components, and the country specific, region specific, and global factors are assumed to be stationary, normally distributed autoregressive processes of order 1.

As the factors and the loadings are not separately identified, to achieve identification we follow Moench, Ng, and Potter (2013) and assume i) that $\Lambda_{\bullet}(L) = \Lambda_{\bullet}$ is a constant lower triangular matrix, with elements having fixed signs on the diagonal, and ii) that the factors have fixed variances. The former normalization ensures that the sign of the factors is identified. This latter assumption is justified by the fact that the data are standardized to have unit variance.

The fact that the DHFM model features time-varying intercepts can be made explicit

by rewriting it as:

$$\begin{aligned}
Z_{rcnt} &= \alpha_{CR,rcnt} + \Pi_{CR,rc}^n(L) GFU_t + e_{Z_{rcnt}} \\
\alpha_{CR,rcnt} &\equiv \Xi_{CR,rc}^n(L) e_{R_{rt}} + \lambda_{C,rc}^n(L) e_{C_{rct}} \\
\Pi_{CR,rc}^n &\equiv \lambda_{C,rc}^n(L) \Lambda_{R.rc}(L) \Lambda_{GFU.r}(L) \\
\Xi_{CR,rc}^n(L) &\equiv \lambda_{C,rc}^n(L) \Lambda_{R.rc}(L)
\end{aligned}$$

where $\alpha_{CR,rcnt}$ are time-varying intercepts and $\Pi_{CR,rc}^n$ is a matrix of coefficient restrictions. This way of writing the model puts in evidence that i) the coefficients $\Pi_{CR,rc}^n(L)$ relating the observables to the global financial uncertainty factor GFU are subject to restrictions involving coefficients at different levels of the hierarchical framework, and that ii) time-varying intercepts $\alpha_{CR,rcnt}$ are naturally modeled in this framework.

D: DHFM estimation with unbalanced panels

We estimate the following four-level dynamic factor model:

$$\begin{aligned}
Z_{rcnt} &= \lambda_{C,rc}^n(L) C_{rct} + e_{Z_{rcnt}} \\
C_{rct} &= \Lambda_{R.rc}(L) R_{rt} + e_{C_{rct}} \\
R_{rt} &= \Lambda_{GFU.r}(L) GFU_t + e_{R_{rt}} \\
\Psi_{GFU}(L) GFU_t &= e_{GFU_t}
\end{aligned}$$

where Z_{rcnt} is the monthly volatility at time t for variable n and country c belonging to region r , with zero mean and standardized to have unit variance. GFU_t denotes the common factor, $\Lambda_{GFU.r}(L)$ is the distributed lag of loadings on the common factor, $e_{R_{rt}}$ is the region-specific variation, R_{rt} is the $(k_R \times 1)$ vector of region-specific factors, where k_R denotes the number of regions, $\Lambda_{R.rc}(L)$ is the distributed lag of loadings on the region-specific factors, $e_{C_{rct}}$ is the country-specific variation, C_{rct} is the $(K_{Cr} \times 1)$ vector of country-specific factors, where K_{Cr} denotes the number of countries in region r , $\lambda_{C,rc}^n(L)$ is the distributed lag of loadings on the country-specific factors, and $e_{Z_{rcnt}}$ is the series specific variation. We then assume that the idiosyncratic components, the country-specific, the region-specific, and the global factors are stationary, normally distributed autoregressive processes of order $q_{Z_{rcn}}, q_{C_{rc}}, q_{R_r}$, and q_{GFU}

$$\begin{aligned}
\psi_{GFU}(L) GFU_t &= \epsilon_{GFU,t}, \quad \epsilon_{GFU,t} \sim N(0, \sigma_{GFU}^2) \\
\psi_{R_{rt}}(L) e_{R_{rt}} &= \epsilon_{R_{rt}}, \quad \epsilon_{R_{rt}} \sim N(0, \sigma_{R_r}^2), \quad r = 1, \dots, k_R \\
\psi_{C_{rc}}(L) e_{C_{rct}} &= \epsilon_{C_{rct}}, \quad \epsilon_{C_{rct}} \sim N(0, \sigma_{C_{rc}}^2), \quad c = 1, \dots, k_{rC} \\
\psi_{Z_{rcnt}}(L) e_{Z_{rcnt}} &= \epsilon_{Z_{rcnt}}, \quad \epsilon_{Z_{rcnt}} \sim N(0, \sigma_{Z_{rcn}}^2), \quad n = 1, \dots, k_{rCN}
\end{aligned}$$

where k_R is the number of regional blocks, k_{rC} is the number of countries in region r , k_{rcN} is the number of series for country c belonging to region r . In principle, the lag order can differ across individual series, country sub-blocks, and regional blocks. Our estimation follows the MCMC algorithm proposed by Moench, Ng, and Potter (2013), and generalizes it to allow for missing observations and unbalanced datasets. The estimation steps are the following:

Let $\mathbf{\Lambda} = (\Lambda_C, \Lambda_R, \Lambda_{GFU})$, $\mathbf{\Psi} = (\Psi_{GFU}, \Psi_R, \Psi_C, \Psi_Z)$, and $\mathbf{\Sigma} = (\Sigma_{GFU}, \Sigma_R, \Sigma_C, \Sigma_Z)$. The steps of the estimation procedure are as follows:

1. Organize the data into blocks (regions) and sub-blocks (countries) to yield Z_{rct} , with $r = 1, \dots, k_R$, $c = 1, \dots, k_{rc}$. Get initial values for $\{C_t\}, \{R_t\}, \{GFU_t\}$ using the EM algorithm by Banbura and Modugno (2014). Use these to produce initial values for $\mathbf{\Lambda}$, $\mathbf{\Psi}$, and $\mathbf{\Sigma}$.
2. Conditional on $\mathbf{\Lambda}$, $\mathbf{\Psi}$, $\mathbf{\Sigma}$, $\{R_{rt}\}$ and the data Z_{rct} , draw $\{C_{rct}\} \forall r \forall c$.
3. Conditional on $\mathbf{\Lambda}$, $\mathbf{\Psi}$, $\mathbf{\Sigma}$, $\{C_{rt}\}$ and $\{GFU_t\}$, draw $\{R_{rt}\} \forall r$.
4. Conditional on $\mathbf{\Lambda}$, $\mathbf{\Psi}$, $\mathbf{\Sigma}$, and $\{R_t\}$, draw $\{GFU_t\}$.
5. Conditional on $\{GFU_t\}, \{R_t\}$, and $\{C_t\}$ draw $\mathbf{\Lambda}$, $\mathbf{\Psi}$, $\mathbf{\Sigma}$.
6. Return to 2.

To take into account the dependence among the factors at different levels, we adopt the same Kalman filter algorithm as in Moench, Ng, and Potter (2013). Following Moench, Ng, and Potter (2013), we assume a standard normal distribution as prior distribution for all factor loadings $\mathbf{\Lambda}$ and for all autocorrelation coefficients $\mathbf{\Psi}$, and an inverse chi-square distribution with 4 degrees of freedom and scale parameter equal to $\sqrt{0.01}$ for the variance parameters.

For our baseline scenario, after discarding the first 15,000 as burn-in, we take another 15,000 draws, storing every fifteenth, so that our results are obtained on the stored 1,000 draws.

E: Comparison with alternative measures of GFU

To assess whether controlling for the block-level variations is important, we compare the information carried by the "global" factor estimated by principal components $\tilde{F}_t$ with the GFU factor extracted using our DHFM model. We proceed as follows. First, we regress the principal component $\tilde{F}_t$ on GFU_t , and save the residuals $\tilde{e}_t$. These residuals can be interpreted as variations that are considered common when the global factor is estimated via principal components, but not when it is estimated using the hierarchical

model. Then, we regress $\tilde{e}_t$ on each regional factor to check whether these residuals can be explained by our estimated regional components. Table A1 reports the estimated coefficient, $\hat{\beta}_{rt}$, and the associated p-value for each of the five estimated model. We find that the North America, Oceania, and Asia block specific factors are all significant, at any conventional level. When we investigate in a similar way the role of country-specific factors, again we find evidence that also country specific factors explain these residuals (results not reported here but available upon request). We interpret this evidence as supportive of the statistical relevance of controlling for region and country specific dynamics to correctly quantify the global factor.

Obviously, there are many alternatives to the hierarchical structure we focus on in the paper. One potential concern regards the dependence of our estimated GFU factor on such a structure. We check the reliability of our GFU measure by constructed four alternative GFU measures that do not depend on the blocks and sub-blocks we work with. First, we construct another proxy of GFU by computing a weighted average of the financial volatility series in our dataset. As weights, we use the (time-varying) share of real GDP of each country to weight the country-specific financial volatility series.¹ As a second alternative to our estimated GFU measure, we compute the common factor across our financial volatility series via a principal component approach (PCA).

There are also alternative ways of computing volatility. In an influential paper, Barndorff-Nielsen and Shephard (2004) propose a model-free measure of realized volatility, realized bipower variation. This way of measuring volatility is quite appealing in our case, as it is shown to be an asymptotically unbiased, model-free econometrics estimator of integrated variance in stochastic volatility models, which is robust to the presence of rare jumps. Using the same notation as in Section 3.1, the month- t measure of realized bipower variation (RBV) for the returns of variable n , ret_{rncntd} , belonging to country c in region r , and observed on day d , is given by

$$RBV_{rncnt} = \sum_{d=1}^{M_{rncnt}-1} \frac{1}{M_{rncnt} - 1} |ret_{rncntd}| |ret_{rncnt(d+1)}| \quad (2)$$

where $|x|$ denote the absolute value of x . We then use the same DHFM as in (??) to get a new measure of Global Financial Uncertainty, based on (2).

¹We use the (time-varying) share of real GDP of each country with respect to the total level of real GDP produced by the 42 countries belonging to our dataset as weights. We constructed the time-varying weights used in this exercise by working with the figures for the real GDP provided by the World Bank, which are available here: <https://data.worldbank.org/indicator/NY.GDP.MKTP.PP.KD>. To compute the GDP shares we used as weights, we divided each country's real GDP figure by the sum of the real GDP of the 42 countries in our panel in each given year of our sample.

As discussed in the paper, to preserve feasibility in estimating our largely parameterized dynamic hierarchical factor model, we assume that the idiosyncratic components are normally distributed. To get closer, at least partially, to the Gaussianity assumption, we estimate our GFU factors from log-volatilities. Figure A3 plots our benchmark GFU measure and these two alternative ones. A few remarks are in order. The GFU index based on realized bipower volatility is less volatile than our baseline GFU. If we use 2 standard deviations as a cut off line to identify large jumps in global financial uncertainty, we find that using RBV only two episodes are identified as global financial "shocks": the GFC and the Covid-19 outbreak, while using our baseline measure of financial uncertainty we find that also the Asian crisis in 1998 and the Euro area sovereign debt crisis in 2011 would be classified as global financial uncertainty "shocks". Focusing on log-volatilities, we notice that it is highly correlated with the baseline GFU measure (0.88). The GFU index based on log-volatilities is more volatile, and displays two major spikes: one corresponding to the GFC and another to the Covid-19 outbreak.

Overall, the checks in this Section of the Appendix show that the different common volatility factors have a very similar temporal evolution and are highly correlated (correlation coefficients with our benchmark GFU factor range from 0.58 (bipower volatilities) to 0.94 (PCA)). We conclude that our benchmark GFU measure is not heavily dependent on our assumptions regarding the composition of the regional sub-blocks in our model nor on the way we have computed monthly volatilities

F: Regional and country-specific factors

Even though the focus of our analysis is the estimation of a global financial uncertainty index obtained after controlling for comovements that are specific to a given region or country, a by-product of the DHFM are region- and country-specific uncertainty measures. In this section we report some selected regional and country specific uncertainty indices, discuss their dynamics, and provide a comparison with the global uncertainty measure. Figure A4 plots each of the regional components against the global index. It is worth noticing that the regional uncertainty indices follow heterogeneous dynamics: while uncertainty in some regions (e.g. Europe and Oceania) display high correlation with the global component, others (Latin America, in particular) show a more prominent idiosyncratic behavior. Prominent examples are: the 2011 sovereign debt crisis in Europe which determined a large increase in uncertainty both at a regional and at a global level, and as such can be classified as a "global" event in terms of financial

uncertainty; the 1997 Asian financial crisis, which has triggered an increase in both the regional and the global component, but with a relatively stronger impact on the regional one; the Latin American currency crises of the early 1990s, which are reflected in a remarkable increase of financial uncertainty within the region, but triggered no noticeable spike at a global level.

A similar heterogenous pattern is evident from the country specific components (see Figures A5 and A6). While uncertainty in some countries (e.g. the US, the UK, Germany, Australia) is highly correlated with the global component, in some other countries (Argentina and China, in particular) financial uncertainty displays barely any correlation with the global component. Telling examples of this heterogenous dynamics are: the 1992 exchange rate crisis in Italy and the 2011 sovereign debt crisis in Greece, which are both events that triggered an increase in uncertainty at a global level but not as much as at a country-specific level; the 1994 Mexican currency crisis, the 2001 debt default in Argentina, and the Chinese economic reforms of the early 1990s, which are all events that triggered enormous spikes in financial uncertainty at a country level, but barely any change in global uncertainty.²

G: VAR analysis: Data and IRFs

Figure A7 plots the data used in our VAR analysis. Figure A8 depicts the median realization of the GFU shocks across the $K = 500,000$ rotations and the dates for the narrative identification restrictions identified with the algorithm explained in the text. Figure A9 plots the impulse responses to the three shocks we work with in our empirical analysis.

Regarding the GFU shocks implied by the 406 models that meet our identification restrictions, it is reassuring to know that none of them features persistence at conventional statistical levels according to a standard t-test conducted on the AR(1) OLS estimated coefficient obtained by regressing each time series of GFU shocks on its own lag and a constant. (Newey-West correction of the standard error applied here.)

²An important remark is that, though the main source of heterogeneity among euro area countries is due to the country-specific volatility of stock prices and government bond yields, the exchange rate volatility after the introduction of the euro is not necessarily the same across countries. The reason is the following. Our monthly realized volatilities are constructed starting from daily data. In a given month, data for two different euro area countries are not necessarily available for the very same days. The reason is that data (including exchange rate data) are not reported for public holidays, and public holidays differ among countries.

H: GFU: The role of COVID-19 observations

Figure 1 in the paper reports our estimate of the GFU factor obtained by working with the sample July 1992-May 2020. Obviously, the last three months of this sample (March-May 2020) are associated with extreme realizations of stock market returns and asset returns more in general. These observations could in principle negatively affect the consistency of our GFU estimates, which are obtained with a fixed-coefficient model (although, as shown in Section B of this Appendix, the model can be re-written in order to put in evidence the time-varying intercepts linking our observables with the GFU factor). Figure A10 contrasts our baseline GFU estimate with the one obtained with pre-COVID19 observations, which we used in a previous version of the paper (sample: July 1992-June 2019). Bottom line - in the common sample, the two estimates coincide. Apart from the time-varying intercepts, which could help capture the different volatilities in the last three months of the sample, our explanation for the irrelevance of the COVID-19 observations on the GFU estimate in the pre-COVID19 sample is the data we use, i.e., monthly volatilities that are computed on the basis of daily returns that are corrected by a month-specific mean. In other words, the data processed by the DHFM are constructed by considering a monthly mean of the various returns we use that already accounts for outliers.

I: Role of different seeds

Baumeister and Hamilton (2020) stress that the seed one sets for initializing the algorithm used to draw different rotations of the Cholesky-VAR variance-covariance matrix can play a role when it comes to producing moments such as impulse responses to identified shocks, or peak responses across different retained models, and so on. Figure A11 plots the summary of our main results in the paper conditional on four different seeds. Our results turn out to be pretty robust to changes in the seed set for the simulations that produce our baseline results.

J: Role of Haar measure and orthonormal matrices

The orthonormal $\mathbf{Q}$ matrices we work with to generate candidate impulse vectors that are then assessed against our identification restrictions are drawn from a distribution that is uniform with respect to the Haar measure over the set of such $\mathbf{Q}$ matrices. As stressed by Baumeister and Hamilton (2020), this does not necessarily imply that the moments of interest are also uniformly distributed across the models $\mathbf{B}$ consistent with

the data. Hence, it is important to check how the moments of interest look like when one considers all models consistent with the data before imposing the identification restriction. This way of proceeding naturally provides us with an assessment of the contribution of our restrictions too. Figure A12 plots the moments of interest in absence of the imposition of any of our identification restrictions. The Figure clearly points to the role of our identification restrictions in dramatically narrowing down and tilting the set of impulse responses to the negative territory as far as the responses of the global financial cycle and the world output are concerned. The set of responses of GFU itself substantially shrinks after the imposition of our restrictions.³ The effect of our restrictions on the empirical distribution of peak (negative) responses of world output is also quite noticeable, with the figures reported for the models suggesting a positive response of output to a GFU shock being those of the very last horizon considered. Finally, and interestingly, the scatter plot involving the world output loss and the global financial cycle deterioration actually points to a positive correlation even in absence of our identification restrictions. However, the cloud of points is much more condensed when imposing our restrictions, and the estimated slope of the interpolating straight line is clearly steeper.

K: Further robustness checks

Figure A13 shows the outcome of perturbing our benchmark analysis along two dimensions: a) doubling the number of draws we employ to explore the space of models consistent with the data, i.e., using one million draws (as opposed to the 500,000 draws used in our benchmark analysis); b) lowering the threshold related to the Great Recession we use to identify the set of retained models. The latter exercise is conducted in order to make sure that the contribution of GFU shocks to global output during the Great Recession is a fact and not an artifact due to our requirement of having large shocks (larger than the 75th percentile of the empirical density generated by our 500,000 draws in September 2008) in that period. We then lower the requirement for the GFU shocks in September 2008 to the 50th percentile of the empirical density. As shown in Figure A13, our results are robust to the above described deviations from our benchmark framework.

³The technical restriction requiring that the $\mathbf{B}$ matrices feature positive elements of the main diagonal is left active while producing the plots in the lower panels of Figure A12. This is the reason why, on impact, all responses of GFU to a global financial uncertainty shock are positive.

L: A novel measure of global financial cycle

Figure A14 proposes a novel measure of global financial cycle. We estimated such a measure by considering our dataset of financial returns (as opposed to volatility of financial returns), which covers 42 countries and different assets (stocks, bonds) and also exchange rate returns. Feeding the dynamic factor model presented in Section 2 of the paper with these data, we obtained an estimated global financial conditions factor. As Figure A14 puts in evidence, our novel global financial conditions factor is positively and relatively highly correlated with Miranda-Agrippino and Rey's (2020), the correlation being 0.51. The factor is available at <https://sites.google.com/site/efremcastelnuovo/home>.

To appreciate the impact of a global financial conditions index more consistent - in terms of information it's based upon - with the GFU factor used in our VAR analysis, we re-ran our VAR by replacing Miranda-Agrippino and Rey's (2020) GFC series with our novel global financial index. Figure A15 plots our IRFs. Bottom line: All qualitative messages we document in the paper on the effects of our identified shocks are robust to the employment of this alternative measure of global financial stress.

M: Different lag structure

Figure A16 proposes the impulse responses obtained with a VAR(5), which features a different number of lags with respect to our baseline VAR(3). The choice of 3 lags is suggested by the Schwarz criterion (a very popular selection criterion in the VAR context), while that of 5 lags is supported by the Akaike criterion (which, according to Ivanov and Kilian (2005), may more reliably capture the dynamic lag structure when it comes to monthly VAR exercises like the one conducted in this paper). The evidence presented in Figure A16 - which is basically the same as the one proposed in the paper and based on a VAR(3) - points to the robustness of our impulse responses to different lag structures.

N: Placebo exercise

Is our evidence on the real effects of GFU shocks genuinely due to global shocks, or are we confounding global shocks with shocks specific to a regional area? In other words, how global the global uncertainty shocks are in our exercise? Could the effects triggered by such shocks be replicated by regional uncertainty factors once one focuses on their components that are orthogonal to GFU? We address this question by estimating a non-

recursive structural VAR that jointly models the financial uncertainty factor assigned by our estimated DHFM to Latin America, the global financial cycle by Miranda-Agrippino and Rey (2020), and the world output measure proposed by the OECD and updated by Baumeister and Hamilton (2019). The idea is the following. Latin America is a very specific region, with bouts of uncertainty that are clearly "local" and due to the financial and real situations characterizing the countries in our panel (Argentina and Mexico in the first place). For instance, the spectacular hikes in regional Latin American uncertainty in the early 1990s documented in Figure A4 that are due to issues related to exchange rates (the "Peso problem") and fiscal sustainability (Argentina's default) are much milder in relative term at a global level according to the reading of global financial volatility offered by our GFU measure. If Latin American financial uncertainty (the component orthogonal to GFU) is meant to have an effect mostly at a regional level, its impact on the global financial and real cycles should be moderate at best.

To verify this point, we run a "placebo" test with which we replace the GFU measure in our trivariate VAR with the estimated financial uncertainty regional factor associated with Latin America. Before running the reduced-form VAR, we "purge" the Latin American factor by regressing it over the GFU in order to control for global effects that are likely to have influenced Latin America as well.⁴ Non-recursive identification is achieved by following the same strategy proposed in the baseline analysis for the GFU shocks, i.e., we employ a combination of various restrictions (narrative, sign, ratio) to pin down the exogenous changes in Latin American financial uncertainty. The identification of the dates for the narrative restrictions is achieved in the same "data-driven" way discussed in our main text, i.e., we run simulations without imposing any identification constraints and recover the median realizations of the (Latin American, in this case) financial uncertainty shocks. Then, we isolate extreme jumps of such shocks (those above 2 std), interpret them, and require them to be large for the Latin American shocks to be actually identified.

Table A3 reports the constraints imposed to identify our interpretative models. Narrative restrictions refer to seven dates, whose interpretation is provided.⁵ All narrative

⁴It is important to recall that the DHFM proposed by Moench, Ng, and Potter (2013), which we work with to estimate our uncertainty factors, delivers global, regional, and country-specific factors that are not orthogonal to each other.

⁵Our algorithm suggests that four dates in late 1992 and early 1993 (November 1992, December 1992, January 1993, March 1993) could be informative for the identification of LAFU shocks. At the same time, the more dates one jointly imposes, the less likely is to end up having a non-empty set of retained models. As a compromise, we select the date among the four with the highest value of the median realization of the shock associated with it, i.e., November 1992.

restrictions require rotations to imply shocks in those dates larger than the median of the unconstrained distribution across unconstrained shocks at each selected date.⁶ We add to such restrictions the ratio and sign restrictions already used in our benchmark investigation on the global effects of GFU shocks. We do so because, once again, we want to make as sure as possible that: a) (Latin American) financial uncertainty shocks do not get confounded with first-moment financial shocks and world output shocks; b) financial uncertainty shocks do not get confounded with oil and US monetary policy shocks.

Figure A17 plots the impulse responses to an LAFU shock (size: one standard deviation) obtained with the above-described set-identification strategy, and compares them with the baseline ones (to a GFU shock). The differences in terms of impact of these two shocks to the global financial cycle and world output are evident. A shock to financial uncertainty in Latin America exerts no effect on the global financial cycle, and a moderate (at best) one on global output. Differently, as documented in the main text, a GFU shock exerts a clearly negative effect on both global indicators. This result is also confirmed by a Cholesky-VAR analysis in which either uncertainty measure (either LAFU or GFU) is ordered last. Wrapping up, the conjecture formulated above (on the effects of shocks to the component of LAFU that is orthogonal to GFU that should have a little-to-none effect outside Latin America) is confirmed by our placebo test.

O: Correlation of GFU shocks with proxies for US financial and macroeconomics uncertainty shocks

While our paper focuses on financial uncertainty, financial volatility could be driven by macroeconomic uncertainty shocks. How confident should we be that it is financial uncertainty we are dealing with, and not macroeconomic uncertainty? We see two levels of analysis here, one that regards the indicators *per se*, and the other one related to the shocks. Regarding the former, we just notice that the highest correlation between GFU and the proxies of US and global uncertainty we deal with in Figures 2 and 3 of the paper is the one with the VIX (0.86), which is typically associated with expected financial volatility. To a deeper level, it is of interest to understand if our GFU shocks are likely to be financial uncertainty shocks as opposed to macroeconomic uncertainty shocks. We

⁶The reason why we do not require the shock during the Great Recession to be larger than the 75th percentile of the distribution of unconstrained shocks is that we are working with the purged measure of financial uncertainty in Latin America, i.e., a measure that takes into account the fact that the Great Recession shock is a global (as opposed to regional) shock.

work with the two proxies of uncertainty proposed by Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021) - namely, financial and macroeconomic uncertainty, estimate a bivariate VAR(3) model, and consider the VAR residuals as proxies for, respectively, US financial uncertainty shocks and US macroeconomic uncertainty shocks. Given the role played by the US as "hegemon" on the international financial markets, if GFU shocks are financial (as opposed to macroeconomic) shocks we would expect GFU shocks to be significantly and positively correlated with the proxy for US financial uncertainty shocks and much less with the proxy for US macroeconomic uncertainty shocks. We investigate this conjecture by correlating the GFU shocks implied by each one of our 406 retained models with the two proxies for US uncertainty shocks.

A few empirical findings are worth noticing. First, the median correlation value between our GFU shocks and the proxy for US financial uncertainty shocks is 0.31, as opposed to the 0.20 we get when considering the proxy for the US macroeconomic uncertainty shocks. Second, Figure A18 plots the histograms of the t-stat values related to these correlations.⁷ All 406 models (implied GFU shocks) display a significant correlation at a 5% level with the proxy for US financial uncertainty shocks. Differently, just 43 out of 406 models (10.59%) feature a significant (5% level of significance) correlation with the proxy for US macroeconomic uncertainty. These two facts combined, while not representing a "smoking gun" in favor of our GFU shocks as necessarily being financial uncertainty shocks, offer evidence consistent with GFU shocks being at the very least connected with financial uncertainty.

P: Reduced-form - structural VAR mapping

Reduced-form VAR. The reduced-form finite-order VAR representation reads:

$$\mathbf{X}_t = \sum_{j=1}^p \mathbf{A}_j \mathbf{X}_{t-j} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim (0, \boldsymbol{\Omega}) \quad (3)$$

where $\mathbf{A}_j$ are matrices of coefficients, $\boldsymbol{\eta}_t$ is the vector of error terms whose variance-covariance is $\boldsymbol{\Omega}$, and $\boldsymbol{\Omega} = \mathbf{P}\mathbf{P}'$, where $\mathbf{P}$ is the unique lower-triangular Cholesky factor

⁷These t-stat values refer to the significance of the slope parameter obtained by OLS regressing the proxy for financial (macroeconomic) uncertainty over a constant and the i th realization of GFU shocks, with i ranging from 1 to 406 (the number of models retained by our set-identification strategy). For the computation of the t-stat values, we consider Newey-West standard errors. The measures of financial and macroeconomic uncertainty for the US are the total uncertainty measures estimated by Ludvigson, Ma, and Ng (2021) (forecasting horizon: on month). Such measures are available at <https://www.sydneyludvigson.com/macro-and-financial-uncertainty-indexes>.

with non-negative diagonal elements. The VAR features equation-specific constants and linear trends. We set $p = 3$, as suggested by the Schwarz information criterion.⁸

Identification. We move from the reduced-form VAR (3) to the structural one as follows. First, we assume that the system of contemporaneous relationships mapping reduced form residuals $\boldsymbol{\eta}_t$ and structural shocks $\mathbf{e}_t$ can be described as

$$\boldsymbol{\eta}_t = \mathbf{B}\mathbf{e}_t, \quad \mathbf{e}_t \sim (0, \mathbf{I}_n) \quad (4)$$

where $\mathbf{B}$ is a matrix featuring n^2 elements. Given that the reduced form covariance matrix $\boldsymbol{\Omega}$ features only $n(n+1)/2$ elements, further restrictions have to be imposed to identify the effects of the structural shocks $\mathbf{e}_t$ on the endogenous variables $\mathbf{X}_t$. Without such further restrictions, infinitely many solutions would satisfy the covariance restrictions $\boldsymbol{\Omega} = \mathbf{B}\mathbf{B}'$. We collect a number $K = 500,000$ of these solutions into the set $\mathcal{B} = \{\mathbf{B} = \mathbf{P}\mathbf{Q} : \mathbf{Q} \in \mathcal{O}_n, \text{diag}(\mathbf{B}) \geq 0, \boldsymbol{\Omega} = \mathbf{B}\mathbf{B}'\}$, where $\mathcal{O}_n$ is a set of orthonormal matrices (i.e., $\mathbf{Q}\mathbf{Q}' = \mathbf{I}_n$).⁹ The set $\mathcal{B}$ is constructed as follows. We rotate $\mathbf{B}$ by drawing K random orthogonal matrices $\mathbf{Q}$. Each rotation is performed by drawing a matrix $\mathbf{M}$ from a multivariate normal $\mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ density as in Rubio-Ramírez, Waggoner, and Zha (2010). Then, $\mathbf{Q}$ is taken to be the orthonormal matrix in the $\mathbf{Q}\mathbf{R}$ decomposition of $\mathbf{M}$. Given that $\mathbf{B} = \mathbf{P}\mathbf{Q}$ and $\mathbf{Q}\mathbf{Q}' = \mathbf{I}_n$, the covariance restrictions $\boldsymbol{\Omega} = \mathbf{B}\mathbf{B}'$ are satisfied. Let $\mathbf{e}_t(\mathbf{B}) = \mathbf{B}^{-1}\boldsymbol{\eta}_t$ be the shocks implied by $\mathbf{B} \in \mathcal{B}$ for a given $\boldsymbol{\eta}_t$. Then, K different $\mathbf{B}$ imply K unconstrained $\mathbf{e}_t(\mathbf{B}) = \mathbf{B}^{-1}\boldsymbol{\eta}_t$, $t = 1, \dots, T$.

Intuitively, what we do is to explore the implications of different rotations of the covariance matrix of the estimated residuals in search of different - but equally likely - interpretations of the effects of GFU shocks consistent with the data. While the so-constructed set $\mathcal{B}$ contains K solutions mathematically coherent with equations (3)-(4), not all these solutions are equally interesting from an economic standpoint. We identify the set of admissible solutions $\bar{\mathcal{B}}$ that can be considered as economically sensible conditional on our research question by imposing different types of identification restrictions, i.e., event constraints, external variable constraints, and sign restrictions. Such constraints are presented and discussed in the paper.

⁸Section S of this Appendix shows that our impulse responses are robust to the employment of 5 lags, which is the lag structure suggested by the Akaike information criterion.

⁹Our results are robust to the employment of $K = 1,000,000$.

Q: Non-recursive vs. recursive identification strategy

Figure 4 in the text offers a comparison between our non-recursive identification strategy and a more pragmatic approach that imposes a recursive structure to the system. We work with a VAR in which we order GFU last to control for business cycle and first-moment financial shocks. The figure reveals the following: i) by construction, the GFU shock in this recursive economy exerts a zero on-impact effect on the global financial cycle. Differently, our non-recursive strategy points to a negative on-impact effect according to all our retained models; ii) as a consequence of i), the point-identified impulse response produced by the recursive VAR features a smaller peak (negative) response of GFC than what suggested by most our retained models; iii) the 95% confidence bands associated with our point-identified responses (generated by the recursive VAR) do not contain all our set-identified responses for a non-negligible number of horizons. This evidence lines up with the one by Ludvigson, Ma, and Ng (2021), who label recursive VARs as "inappropriate" when it comes to studying the business cycle effects of uncertainty shocks (in their case, with US data).¹⁰

This point can be further appreciated by considering a forecast error variance decomposition analysis that focuses on "very short-run" fluctuations. We select a 3-month horizon, compute the forecast error variance decomposition for the three shocks in our VAR, and compare the figures implied by our non-recursive identification strategy (median values across the retained models) with those implied by the recursive identification strategy which orders world output first, the global financial cycle second, and GFU last. Table A4 reports the related figures. The non-recursive identification strategy assigns a larger contribution to GFU as far as the forecast error variance of the global financial cycle is concerned (about 28% vs. 19% in the case of the recursive strategy). According to our non-recursive scheme, the short-run contribution of GFU shocks to world output volatility is somewhat low (3.45%), but still much larger than that implied by the Cholesky VAR (1.66%). Due to the ordering of the variable, which puts GFC before GFU, this latter VAR points to a contribution of the global financial cycle shock to the 3-month ahead variance of the forecast error of GFU as large as 8.59%, while our non-recursive identification strategy suggests less than 1%. Somewhat interestingly, output "shocks" are associated with a contribution to the forecast error volatility of GFU of about 10% vs. a mere 0.27% suggested by the recursive identification strategy.

¹⁰Our Appendix shows that the on-impact response of GFU to a GFC shock is also non-zero. Hence, while orderings of the variables alternative to the one proposed in the text are possible, no recursive VAR would be able to properly capture the GFU-GFC contemporaneous interconnection.

Wrapping up, while displaying some similarities in shape and magnitude, the impulse responses produced with our non-recursive identification strategy also suggest a different contribution - at least at a short-horizon - of our identified shocks to the volatility of the global indicators we model with respect to the description offered by a recursive VAR.

R: WIP construction

The OECD original series is available from January 1958 to October 2011. To extend the data set after October 2011, Baumeister and Hamilton (2019) applied the same methodology used by the OCED. Specifically, they used OECD industrial production and industrial production for the individual non-member countries which are available in the MEI database and applied the weights (which are updated on a yearly basis) reported by the OECD to aggregate those series into a single index. The source of the weights data is the International Monetary Fund’s World Economic Outlook (WEO) database.

S: Role of the ratio restriction concerning the first-moment financial shock

Our baseline VAR results are obtained via a set of restrictions, among which a ratio restriction that requires a first-moment financial shock to have a quantitatively larger on-impact effect on the global financial cycle than on global uncertainty. While being intuitive, this restriction may be questionable because changes in volatility, possibly related to macro fundamentals, could be the main driver of larger movements in asset prices (see, e.g., Kim and Kim (2019)). Figure A19 shows that our results are robust to dropping such restriction.

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North America	Europe	South America
Canada	Austria	Argentina
Mexico	Belgium	Brazil
United States	Czech Republic	Chile
	Denmark	Colombia
	Finland	Peru
	France	
Asia	Germany	Oceania
China	Great Britain	Australia
Hong Kong	Greece	New Zealand
India	Hungary	
Indonesia	Ireland	
Japan	Italy	
Korea	Netherlands	
Malaysia	Norway	
Pakistan	Poland	
Philippines	Russia	
Singapore	Spain	
Taiwan	Sweden	
Thailand	Switzerland	
	Turkey	

Table A1: **Countries and regions covered by our dataset.**

Regional factor	$\hat{\beta}_r$	p-value	R^2
North America	0.145	0.000	0.076
Europe	0.008	0.695	0.000
Oceania	0.099	0.001	0.035
Latin America	0.069	0.487	0.000
Asia	-0.045	0.048	0.012

Table A2: **Correlation with regional factors.**

<i>Event restrictions</i>		
t	Event	Restriction
1992M11	Argentina's set of reforms	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
1994M1	Recessionary winds affecting Argentina	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
1995M5	Peso crisis	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
2007M8	Instability on international financial markets	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
2008M5	Uncertainties about asset bubbles and debt solidity	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
2008M9	Banking crisis	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
2010M12	Hikes in commodity prices	$e_{LAFU,t} > P_{50}(\mathbf{e}_{GFU_t}(\mathbf{B}))$
2007M10-2009M6	Great recession	$0 > \sum_{t=2007M12}^{2009M6} e_{WIP,t}$
<i>Correlation restrictions</i>		
	$ \rho(e_{LAFU,t}, e_{USMP_t}) < 0.05$	$ \rho(e_{LAFU,t}, e_{OIL_t}) < 0.05$
<i>Sign restrictions</i>		
	$\frac{\partial WIP_t}{\partial e_{GFC,t}} > 0$	$\frac{\partial GFC_t}{\partial e_{WIP,t}} > 0$
<i>Ratio restrictions</i>		
	$\left \frac{\partial LAFU_t}{\partial e_{GFC,t}} \right < \left \frac{\partial GFC_t}{\partial e_{GFC,t}} \right $	$\left \frac{\partial LAFU_t}{\partial e_{LAFU,t}} \right > \left \frac{\partial GFC_t}{\partial e_{LAFU,t}} \right $

Table A3: **Set of identifying restrictions: Financial Uncertainty shock in Latin America.** Constraints imposed to identify Latin American financial uncertainty and separate them from global financial cycle shocks and global output shocks. LAFU stands for Latin America financial uncertainty; GFC for global financial cycle; WUI for World Industrial Production.

<i>3-month horizon</i>			
Non-recursive ident. strategy			
Variable/shock	e_{GFU}	e_{GFC}	e_{WIP}
GFU	88.64%	0.73%	10.33%
GFC	28.29%	52.82%	19.17%
WIP	3.45%	6.14%	89.73%
Recursive ident. strategy			
Variable/shock	e_{GFU}	e_{GFC}	e_{WIP}
GFU	91.14%	8.59%	0.27%
GFC	18.75%	70.57%	10.68%
WIP	1.66%	5.89%	92.45%

Table A4: **Forecast Error Variance Decomposition: 3-month horizon, comparison with a recursive VAR.** Forecast error variance decomposition conditional on our retained models (set identification) vs. a point-identified model related to a reduce-form VAR with variables ordered as follows: WIP, GFC, GFU. Figures for the non-recursive identification strategy refer to the median value across retained models of the contribution of each given shock for each given variable at selected horizons. Figures for the recursive identification strategy are point estimates. The sum of the contributions of the shocks per each given variable regarding the set-identified effects of the three shocks does not return 100 percent because this Table considers moments of distributions across the different models of our retained set.



Figure A1: **Our dataset: Geographical coverage.** Countries included in our dataset identified in red.

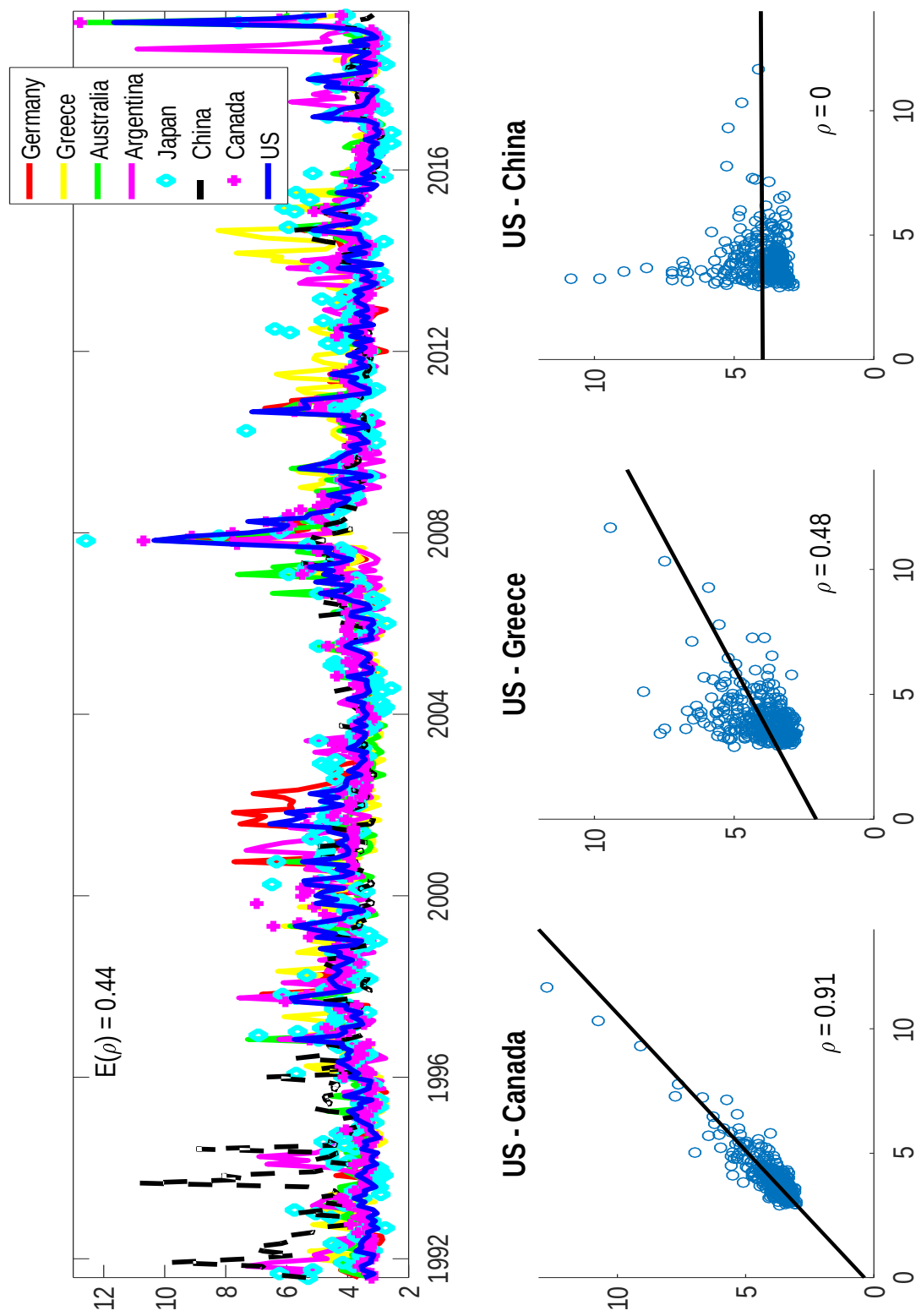


Figure A2: **Financial volatilities around the world.** Upper panel: Stock market returns volatility, selected countries. Lower panels: Scatter plots of stock market returns volatilities, selected pairs of countries.

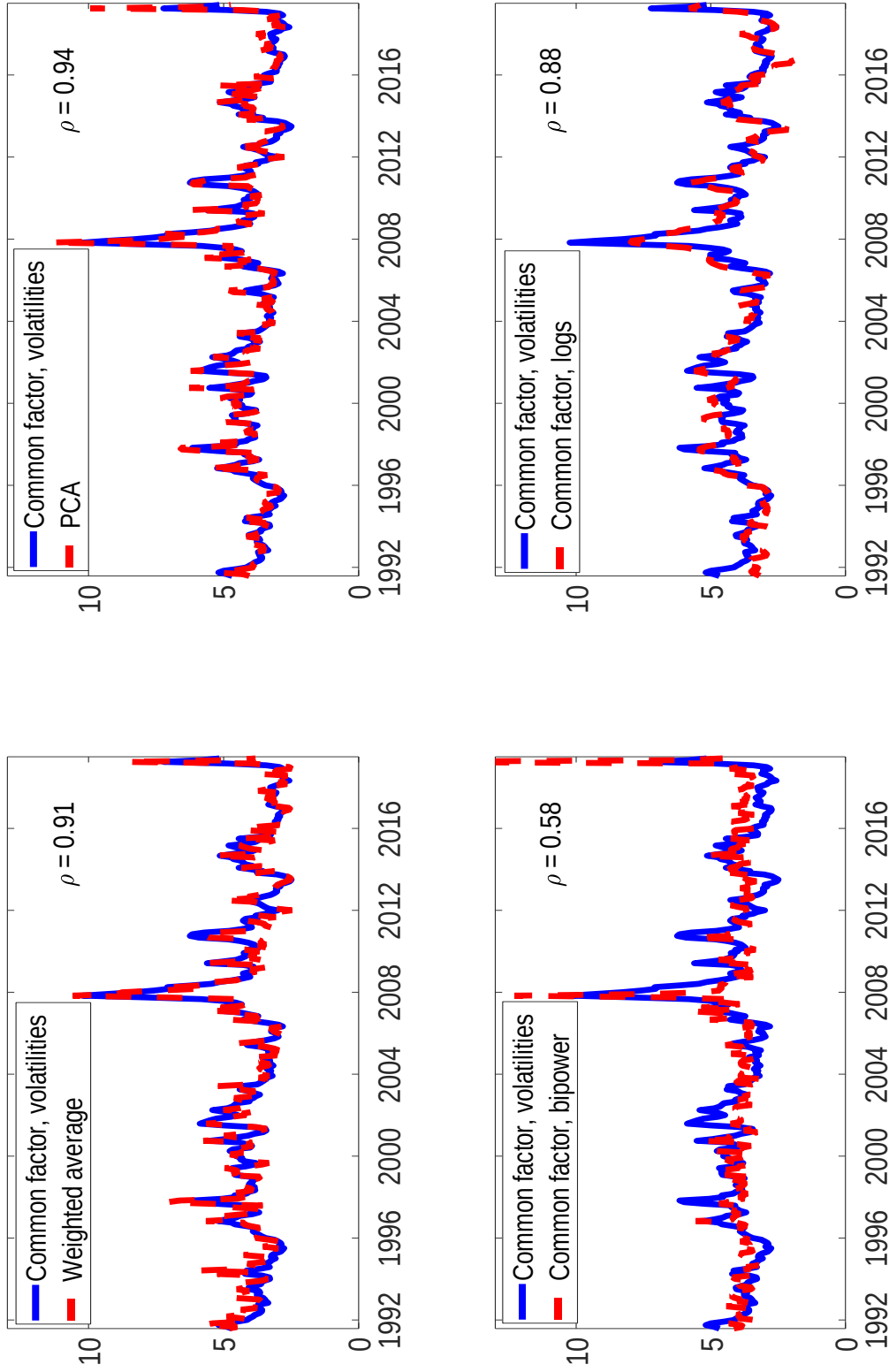


Figure A3: **Measures of global financial uncertainty.** Common factor, volatilities: GFU measure estimated with the dynamic factor model presented in the text. Weighted average: GFU measure computed as a weighted average of the financial volatility measures in our dataset - weights: Time-dependent shares of real GDP conditional on an overall GDP level obtained by summing up the real GDPs of the 42 countries in our dataset. PCA: Common component of our financial volatility series computed with a principal component approach. Common factor, bipower: GFU measure estimated with the dynamic factor model presented in the text and conditional on bipower volatilities computed as in Barndorff-Nielsen and Shephard (2003). Common factor, logs: GFU measure estimated with the dynamic factor model presented in the text and conditional on log-volatilities.

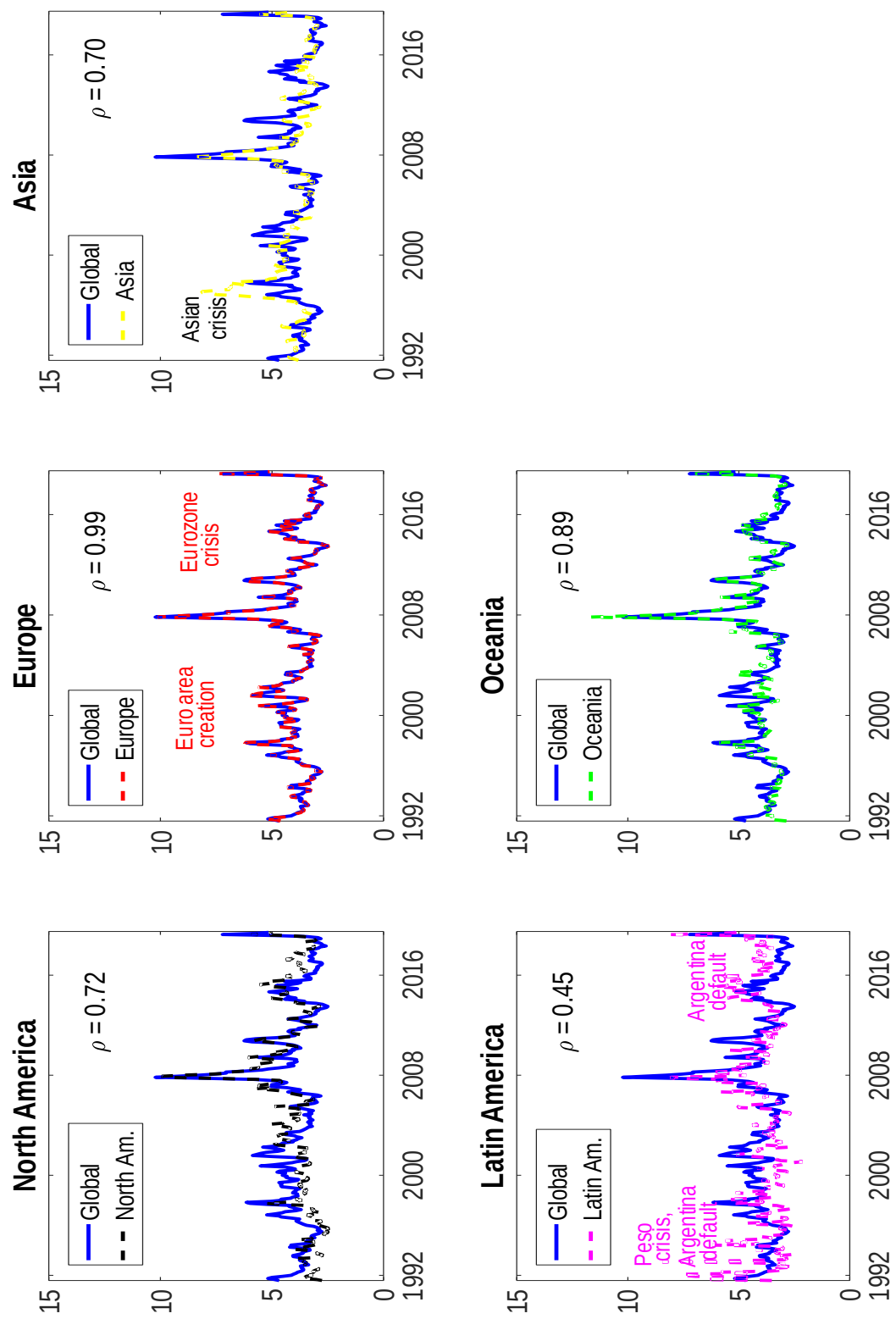


Figure A4: **Regional factors.** Construction of the series explained in the text.

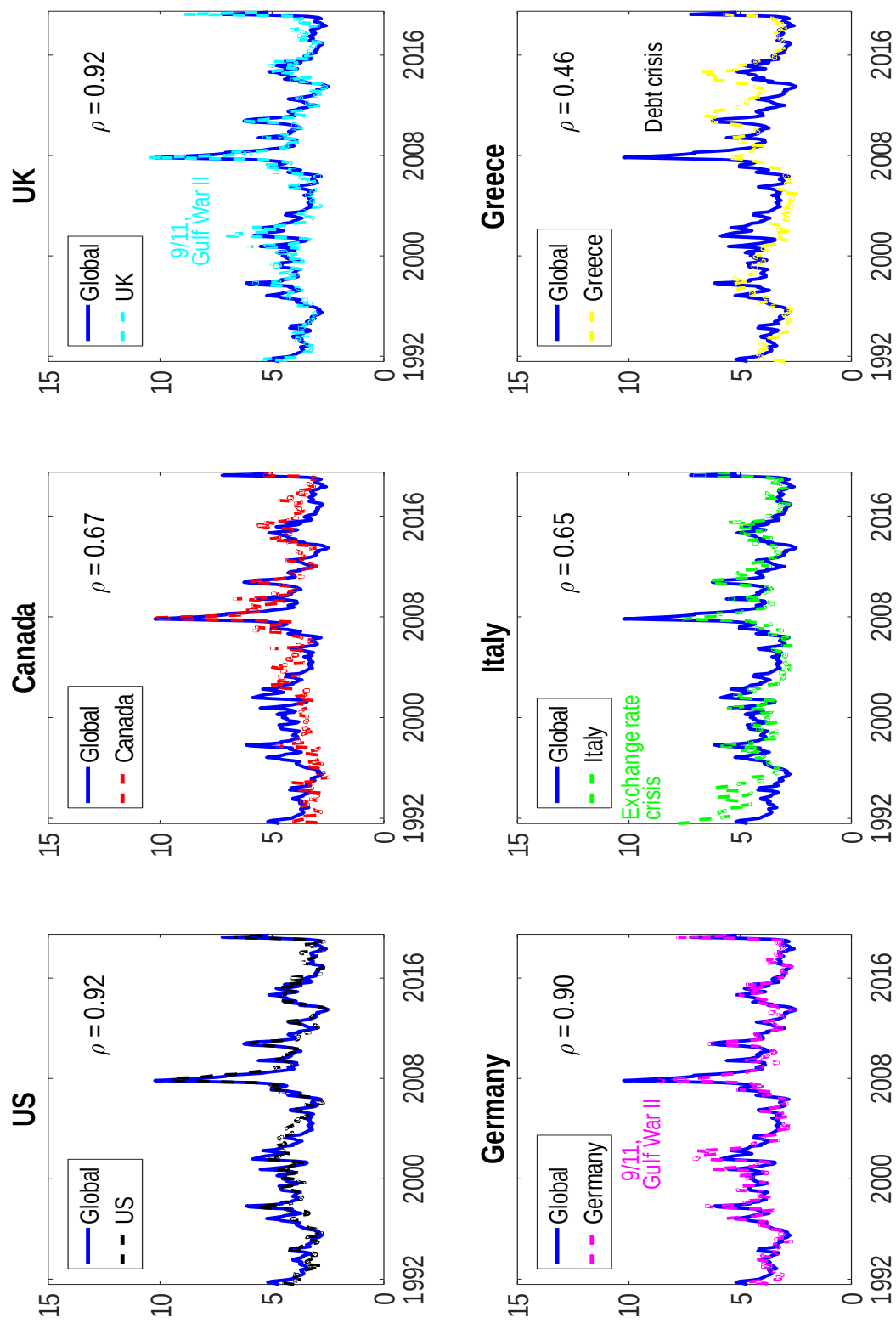


Figure A5: Country-specific factors (selected countries). Construction of the series explained in the text.

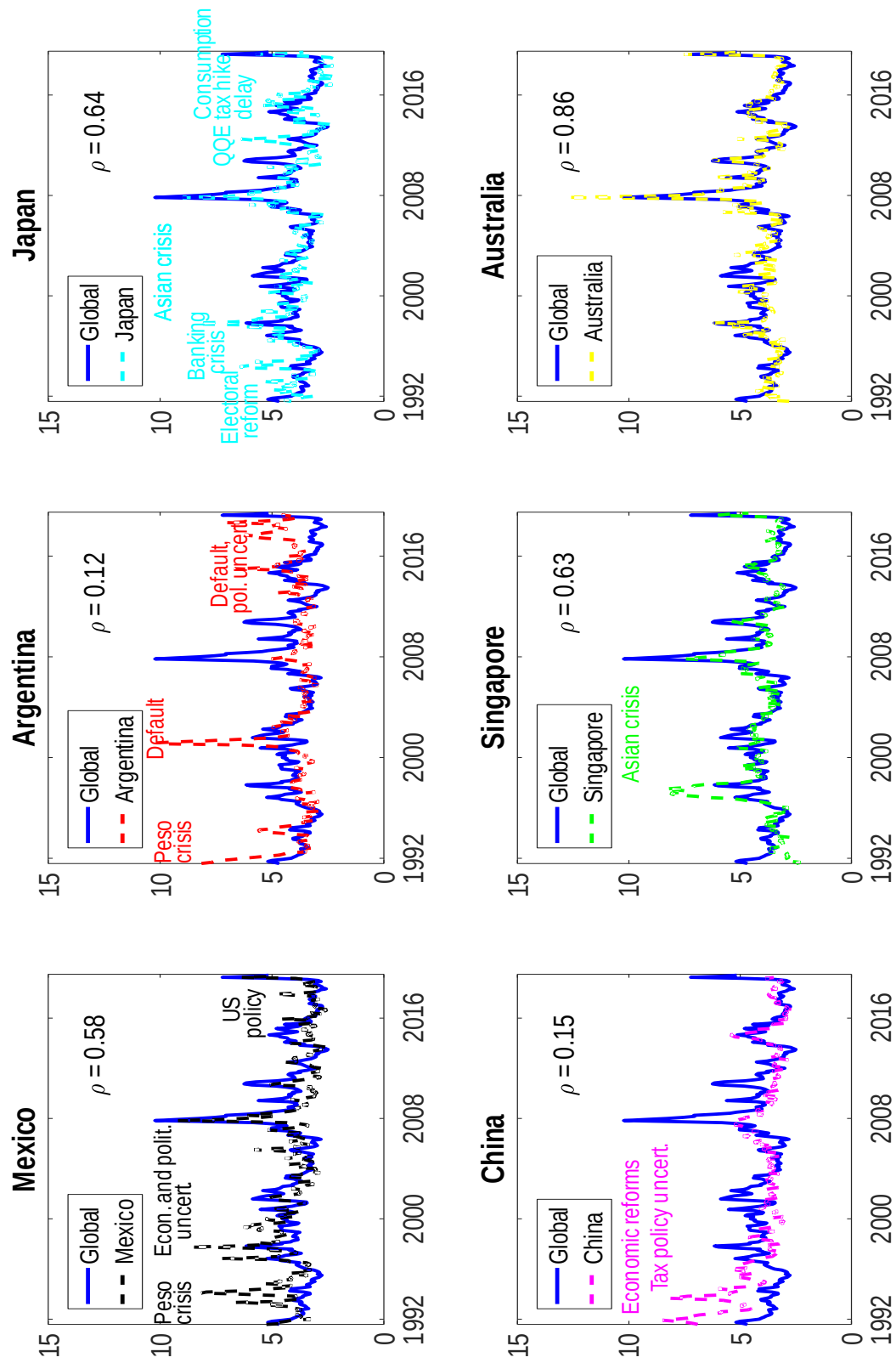


Figure A6: Country-specific factors (selected countries). Construction of the series explained in the text.

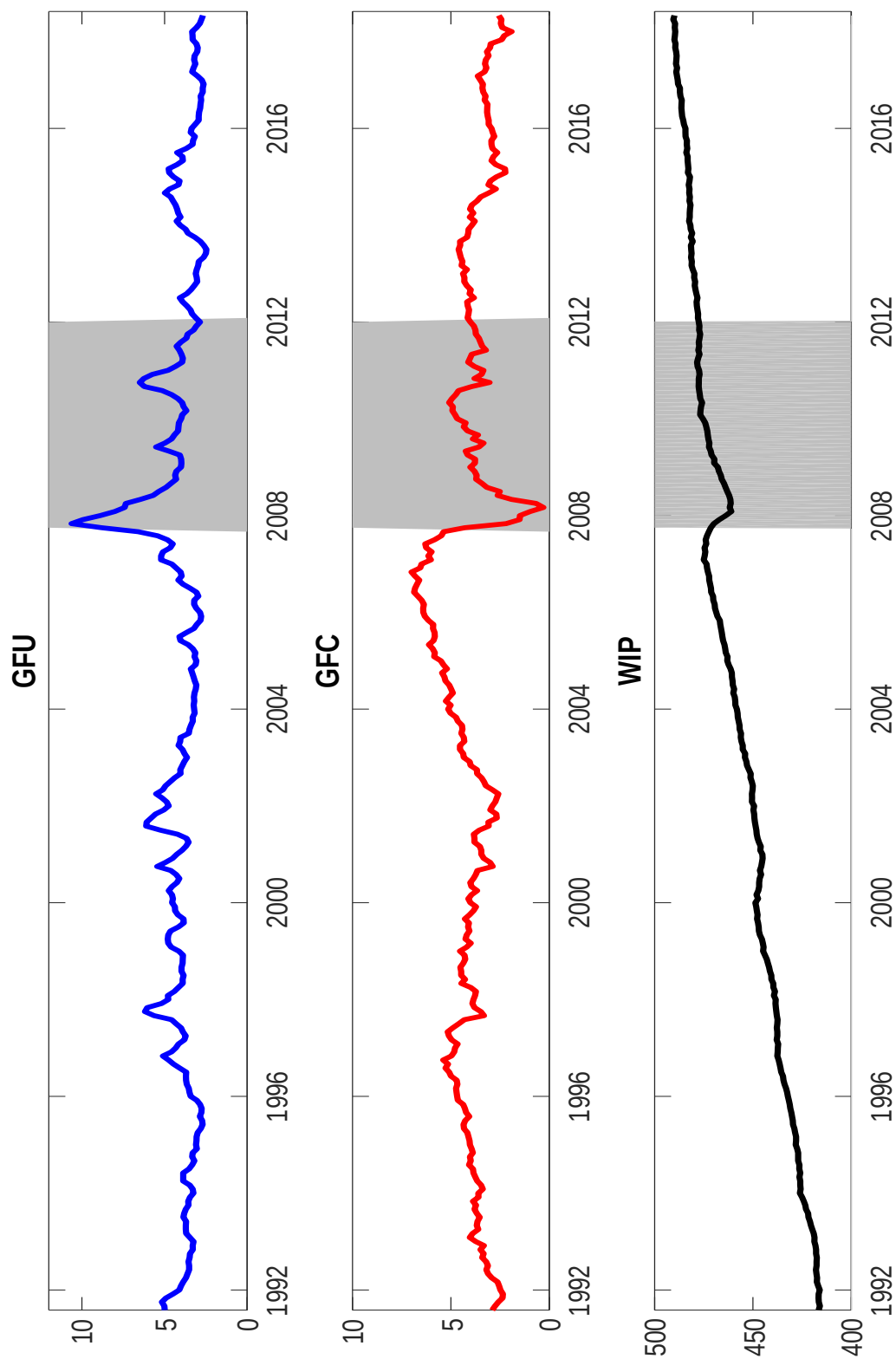


Figure A7: VAR analysis: Series.

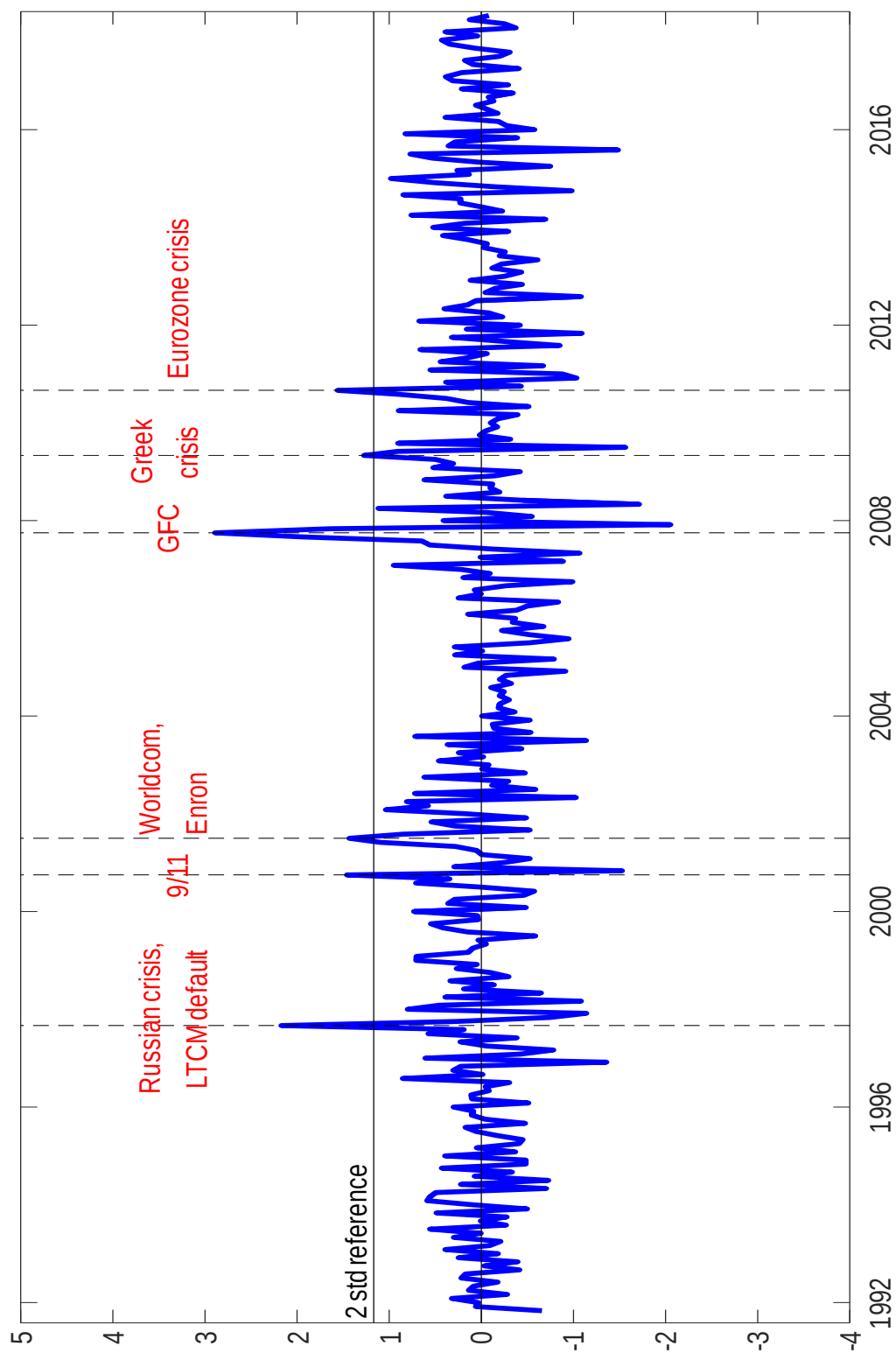


Figure A8: **Unconstrained shocks: Median across realizations.** Dates identified with dashed vertical lines are those corresponding to median values exceeding two standard deviations of the time series of the median realizations itself.

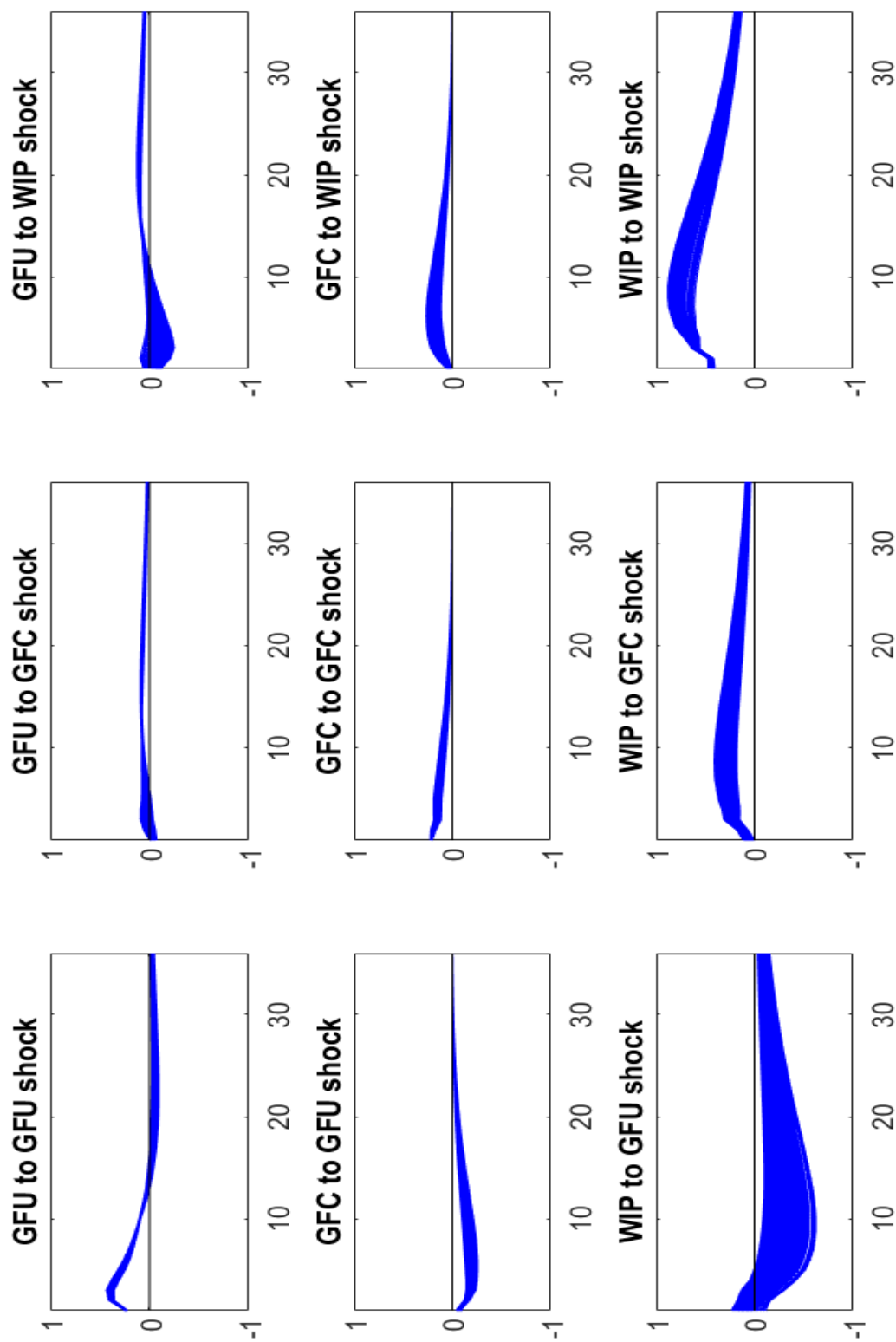


Figure A9: **Impulse Responses to all shocks.** Sample: 1992M7-2012M12. Responses to a one standard deviation shock. Identification of the shock achieved by imposing the mix of event and sign restrictions indicated in Table 1.

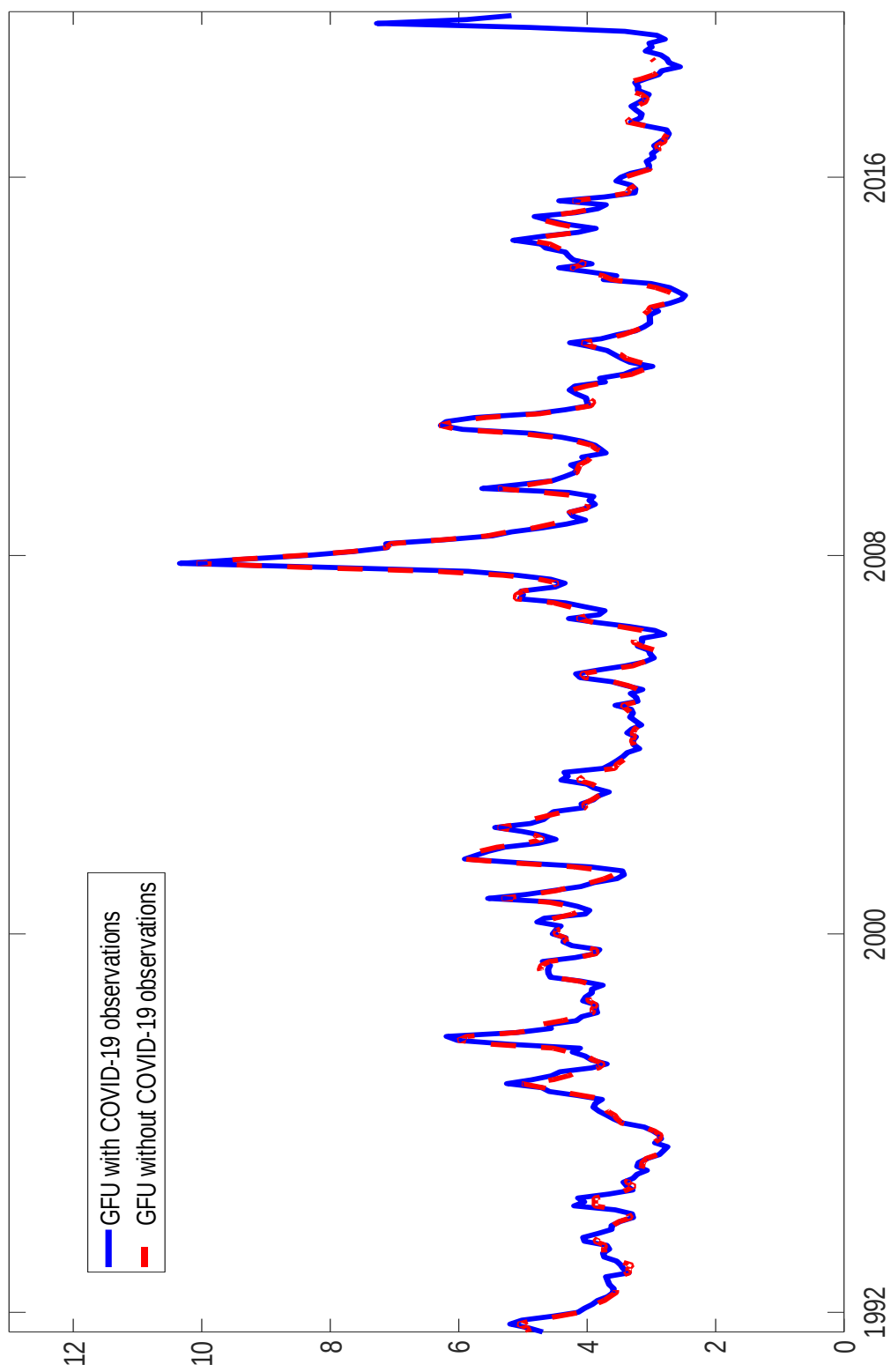


Figure A10: **Global Financial Uncertainty factor: Role of COVID-19 observations.** Factors estimated with the very same dynamic hierarchical factor model over two different samples, i.e., 1992M7-2019M6 (without COVID-19 observations) vs. 1992M7-2020M5 (with COVID-19 observations). Factor standardized over the common sample 1992M7-2019M6.

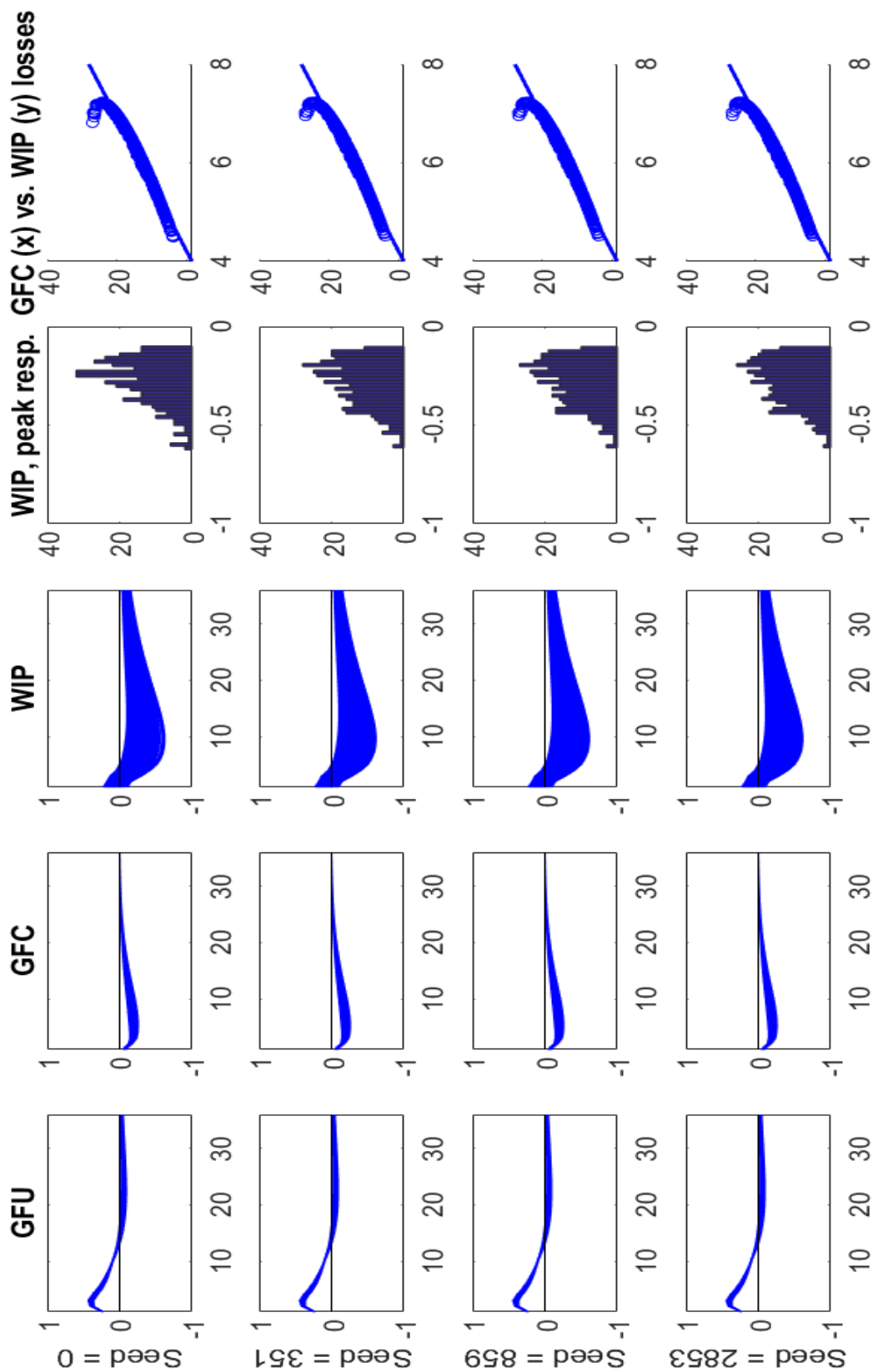


Figure A11: **Macroeconomic Implications of GFU Shocks: Role of Different Seeds.** Impulse responses to a GFU shock (first three columns), peak responses of world industrial production across retained models (fourth column), and scatter plot of WIP loss vs. GFC loss (fifth column) across different seeds initializing our simulations. Seed = 0: Baseline case.

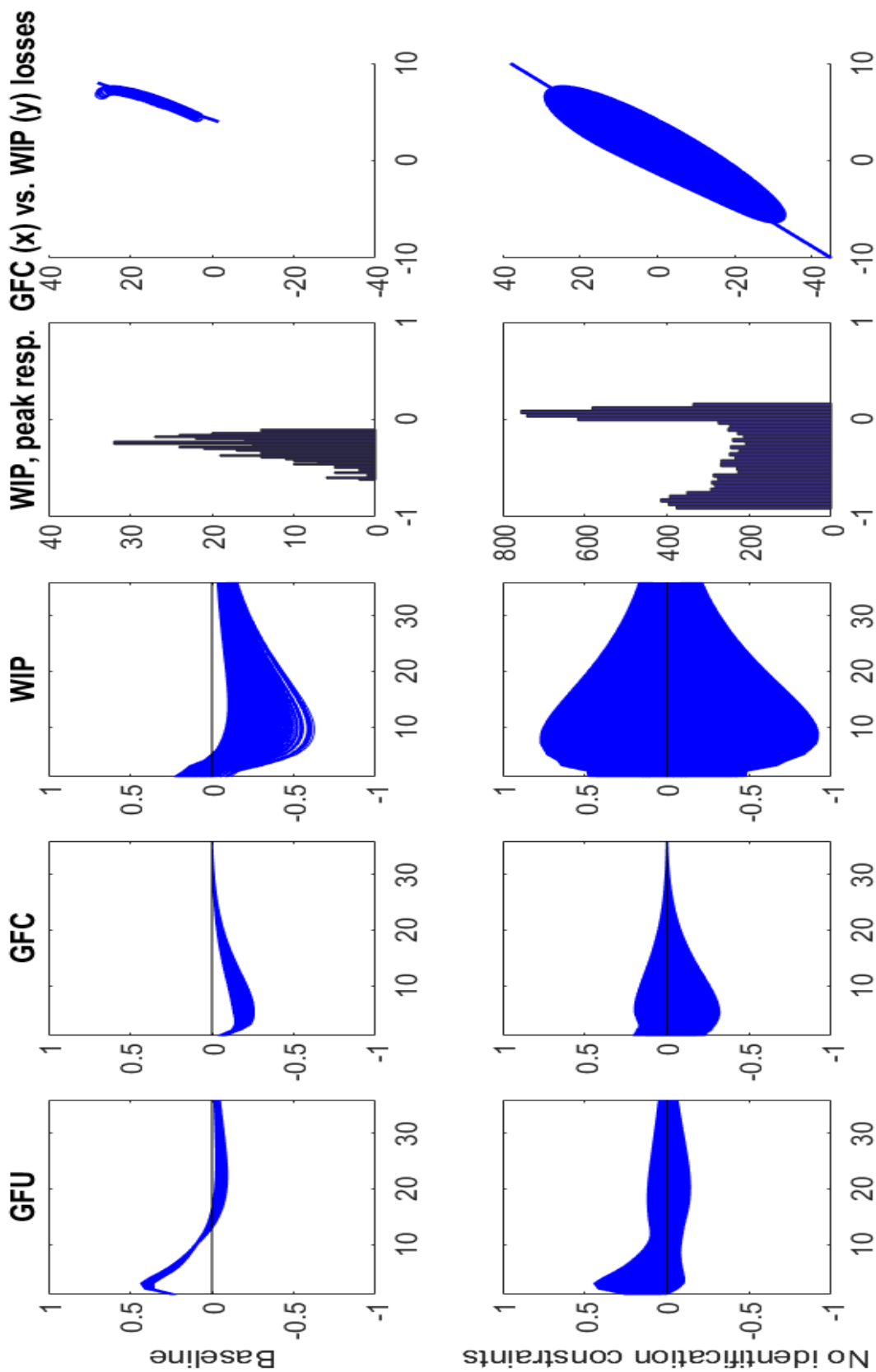


Figure A12: **Impact of the Haar Distribution on Our Results.** Upper panels: Baseline results obtained by imposing the identification restrictions as explained in the text. Lower panels: Moments implied by all rotation matrices without imposing any identification restriction. Moments in the lower panel conditional on just 10,000 (all retained) draws to facilitate the production of the figure.

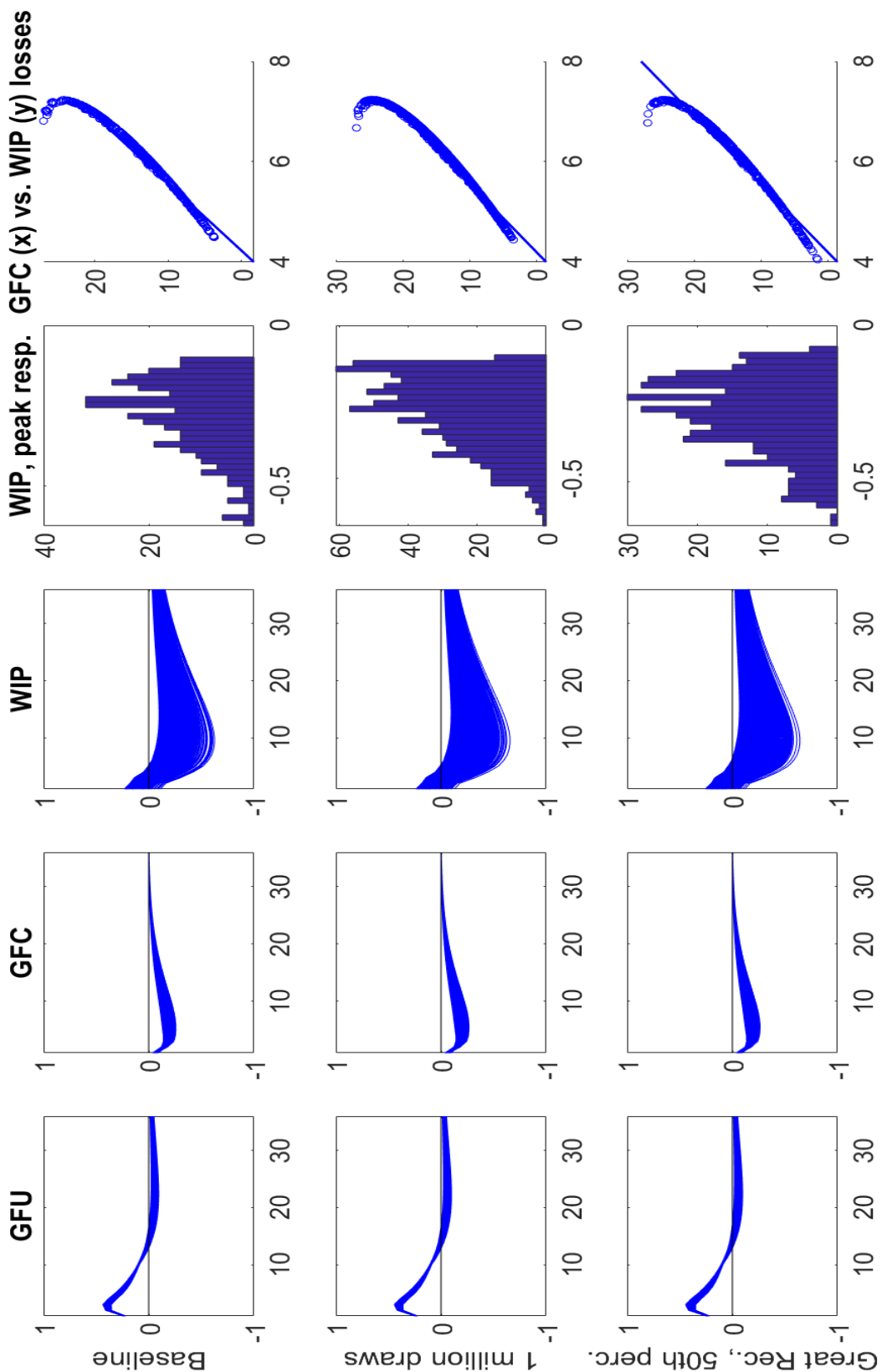


Figure A13: **Macroeconomic Implications of GFU Shocks: Robustness Checks.** Impulse responses to a GFU shock (first three columns), peak responses of world industrial production across retained models (fourth column), and scatter plot of WIP loss vs. GFC loss (fifth column) across three different cases: Baseline (first row); empirical results obtained with one million draws (as opposed to 500,000, as in the baseline case); empirical results obtained by requiring candidate models to generate a GFU shock during the Great Recession larger than the 50th percentile of the distribution of the shocks in 2008M9 (as opposed to the 75th percentile, as in the baseline case).

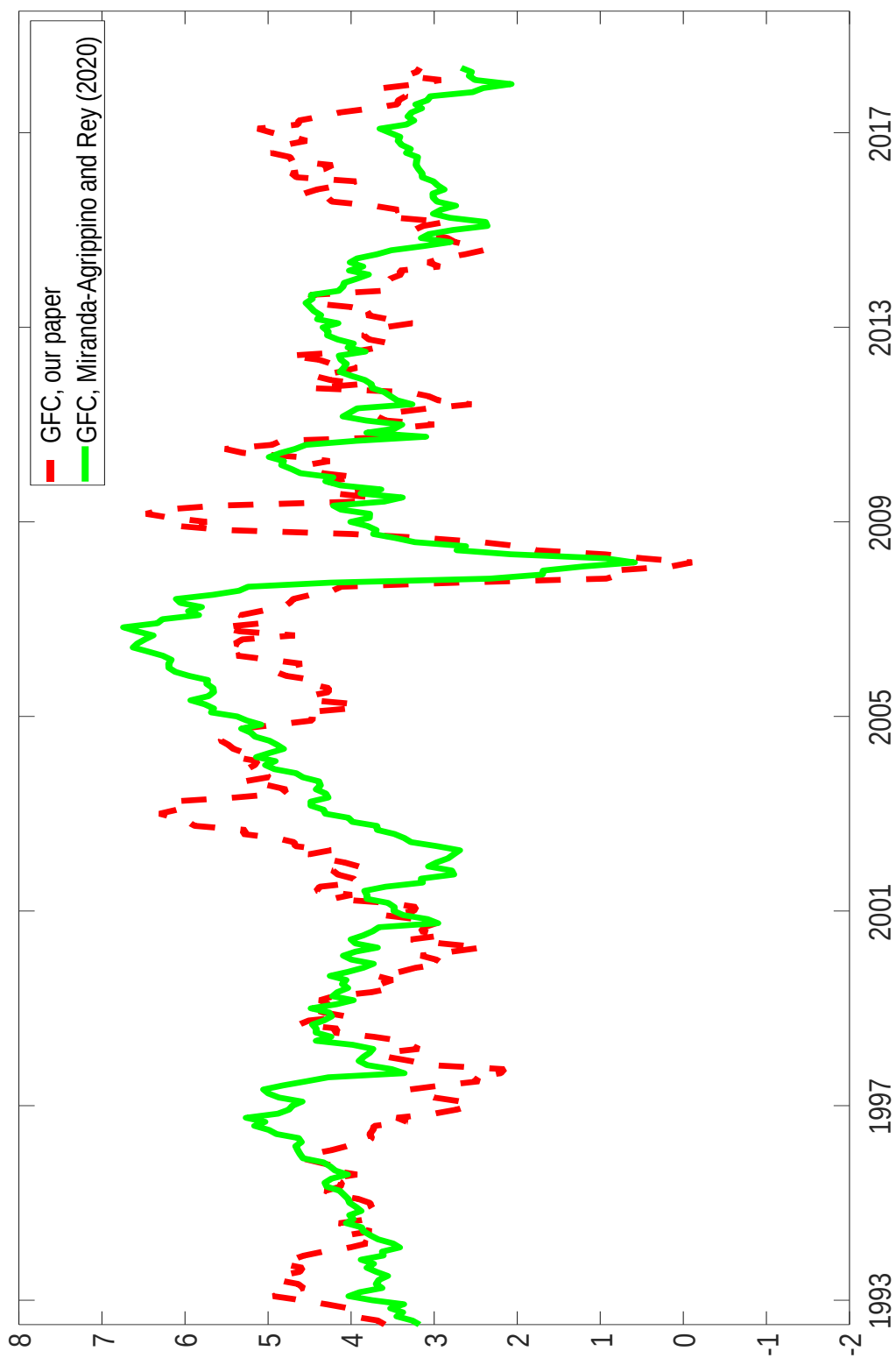


Figure A14: **Global Financial Cycles: Miranda-Agrippino and Rey's (2020) vs. our estimated factor.** Sample: 1992M7-2019M4. Estimation of our global financial cycle factor undertaken by working with the dynamic factor model presented in Section 2 of our paper. The factor is the global factor in such a framework. Data: Financial returns for 42 countries .

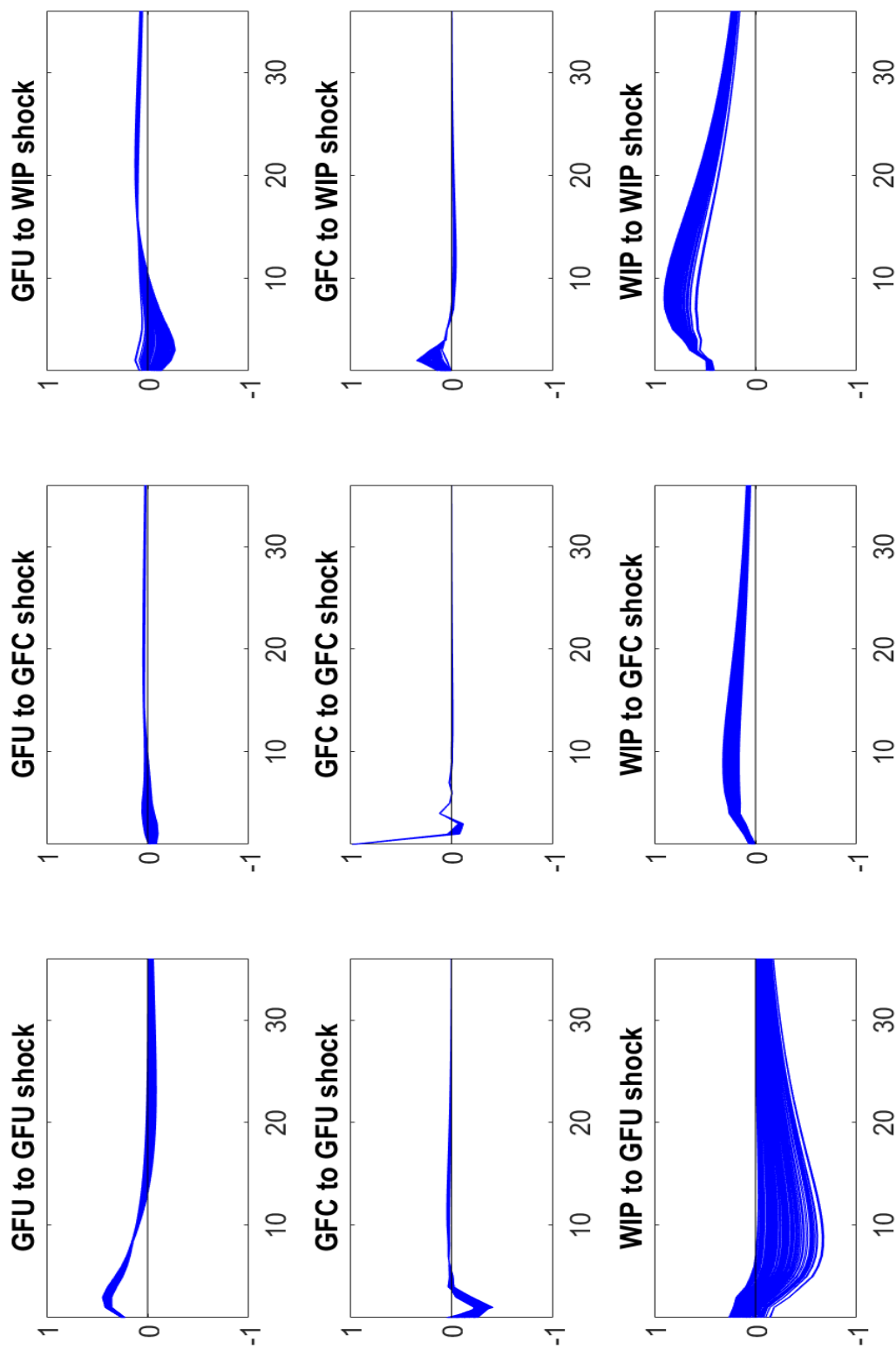


Figure A15: **Impulse Responses to all shocks: VAR featuring our novel measure of global financial cycle.**
Sample: 1992M7-2019M4. Responses to a one standard deviation shock. Identification of the shocks achieved by imposing the mix of event and sign restrictions indicated in Table 1. Measure of global financial cycle estimated as the global component of a dynamic factor model, data: financial returns for 42 countries.

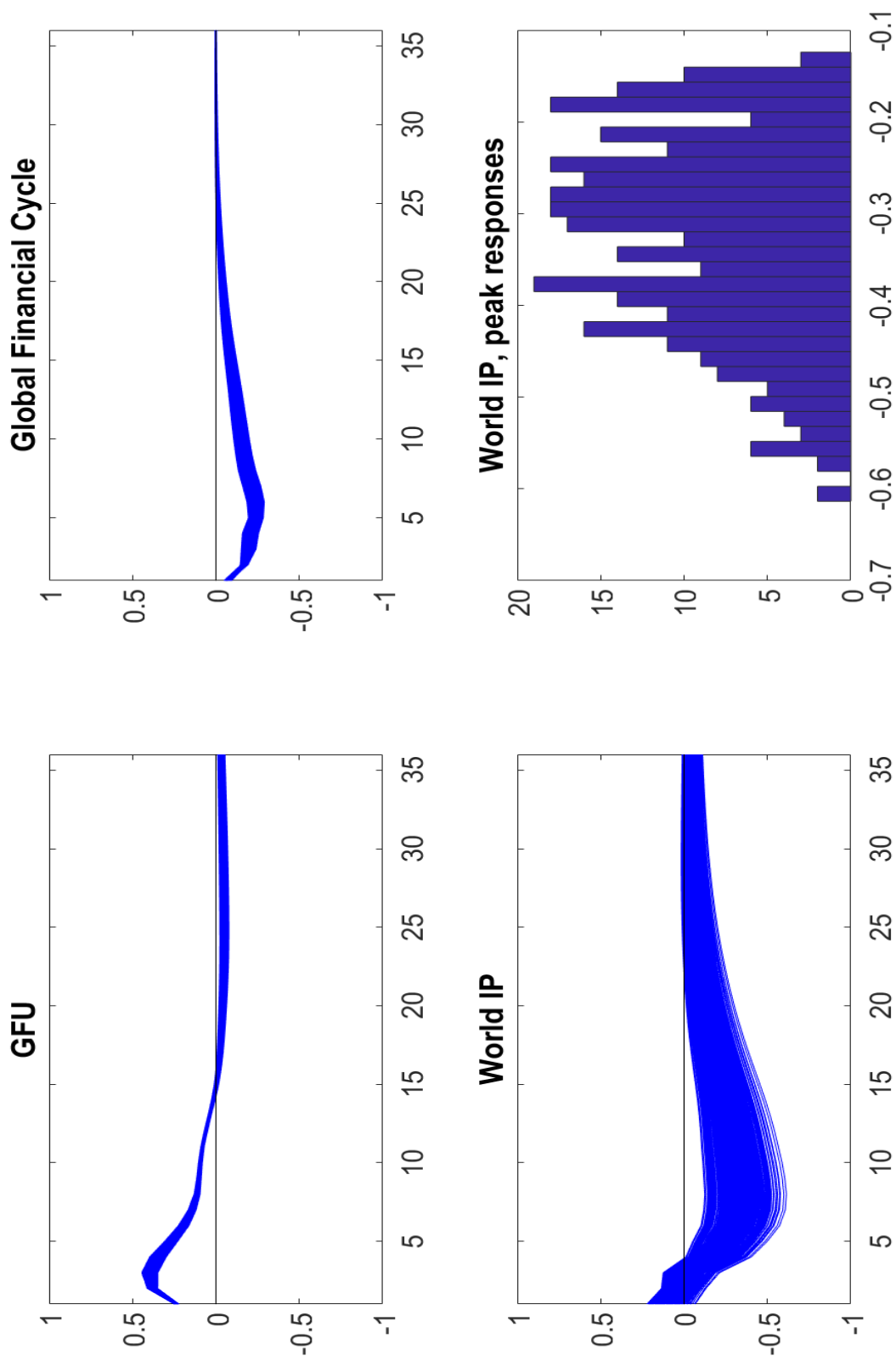


Figure A16: Impulse Responses to a Global Financial Uncertainty Shock: VAR with 5 lags. Sample: 1992M7-2019M4. Blue solid lines: Set-identified responses to a one standard deviation GFU shock. Identification achieved by imposing the restrictions reported in Table 2. Bottom-right panel: Distribution of the peak responses across set-identified models.

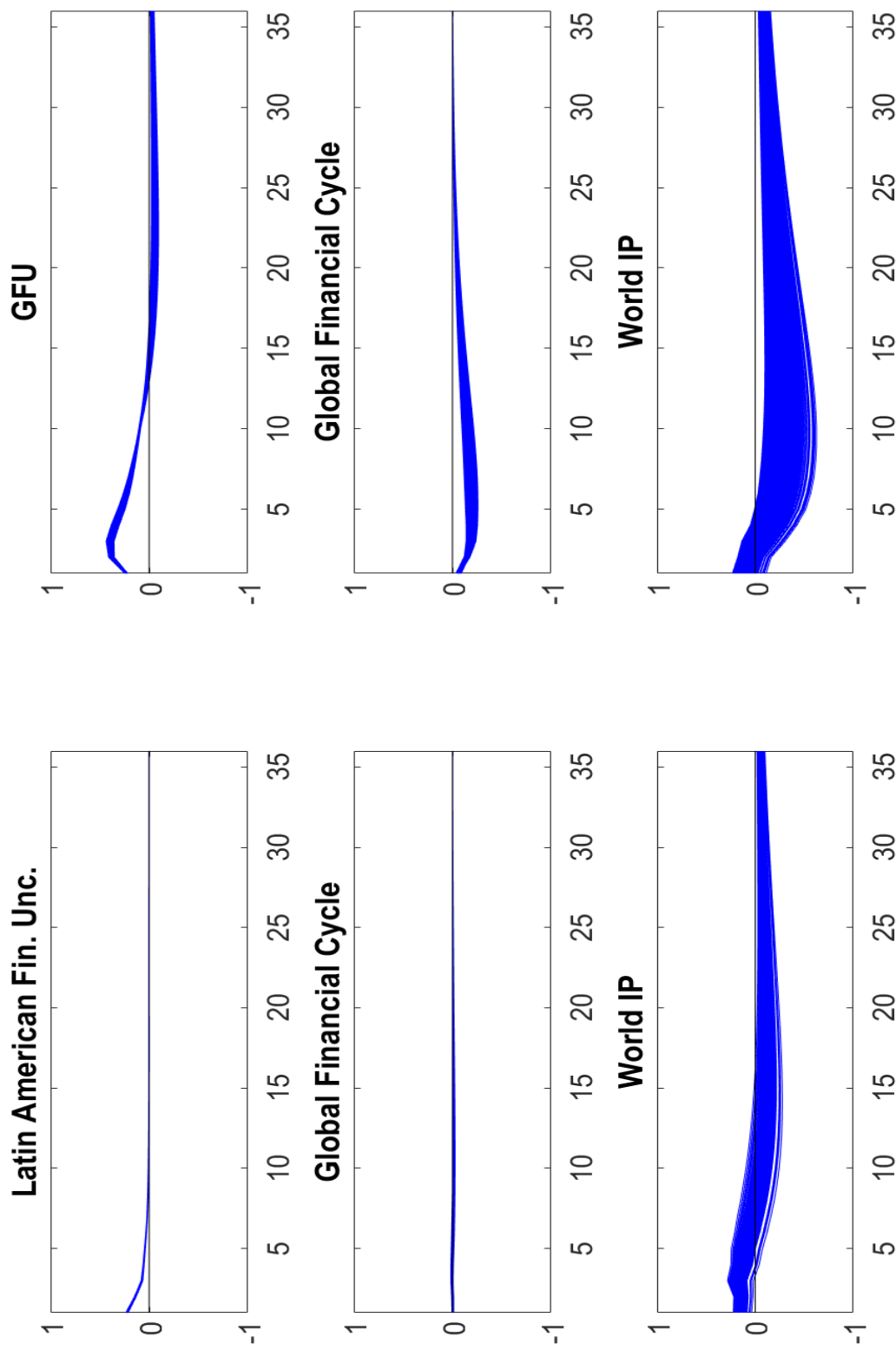


Figure A17: **Impulse Responses to a Latin American Uncertainty Shock.** Sample: 1992M7-2019M4. Left column: Set-identified responses to a one standard deviation shock to Latin American uncertainty. Identification achieved by imposing the restrictions indicated in Table A3 in this Appendix. Right column: Set-identified responses to a one standard deviation GFU shock. Identification achieved by imposing the restrictions reported in Table 2 in the paper.

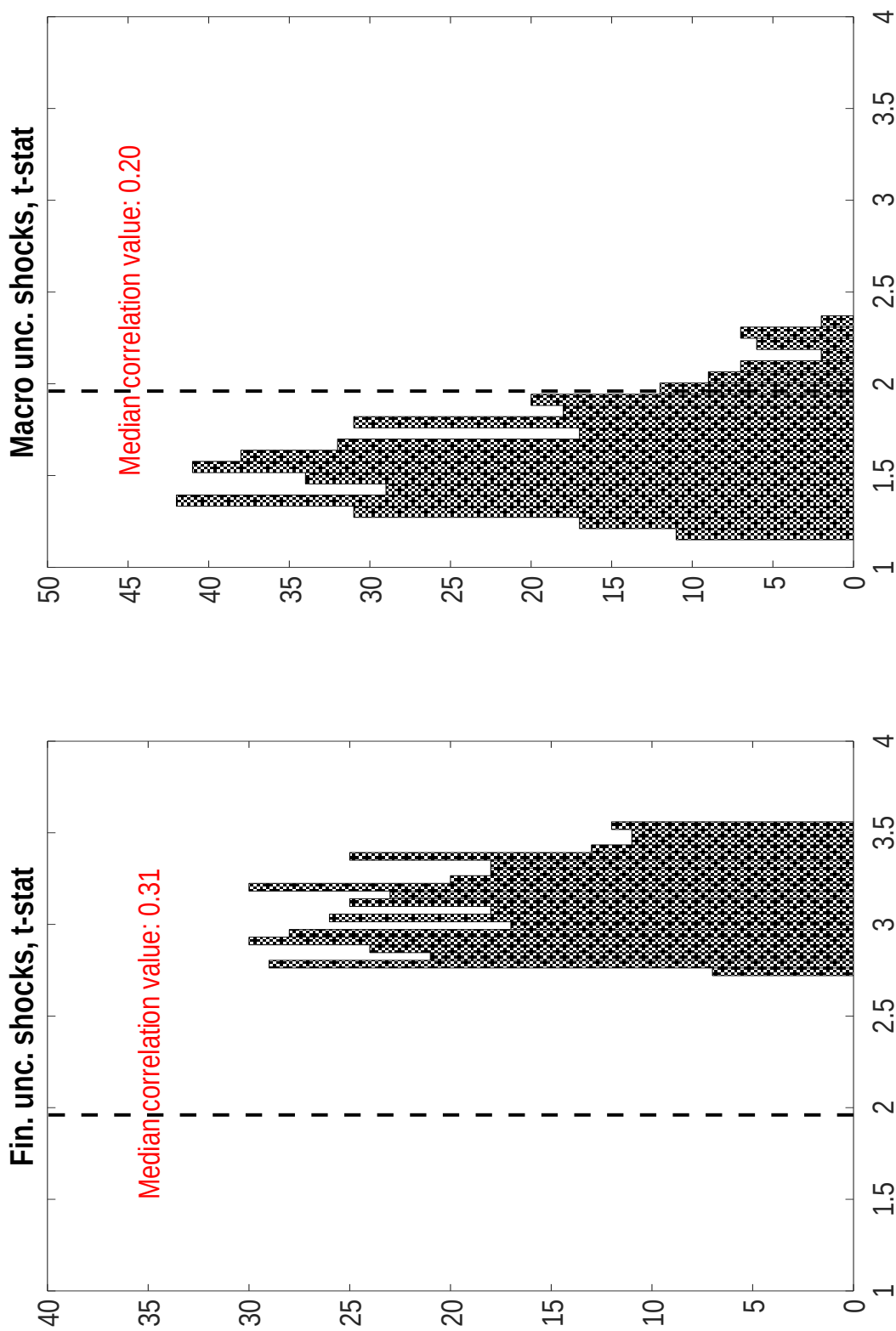


Figure A18: **Correlation between GFU shocks and proxies for financial/macro uncertainty shocks: t-stat values.** Sample: 1992M7-2019M4. Left panel: Empirical distribution of t-stat values of the correlation between GFU shocks and a proxy for financial uncertainty shocks estimated with a bivariate VAR(3) modeling the Ludvigson et al. (2021) financial and macroeconomic uncertainty indicators. Right panel: Empirical distribution of t-stat values of the correlation between GFU shocks and a proxy for macroeconomic uncertainty shocks estimated with a bivariate VAR(3) modeling the Ludvigson et al. (2021) financial and macroeconomic uncertainty indicators. Vertical dashed lines: 1.96 (threshold for 5 percent significance level).

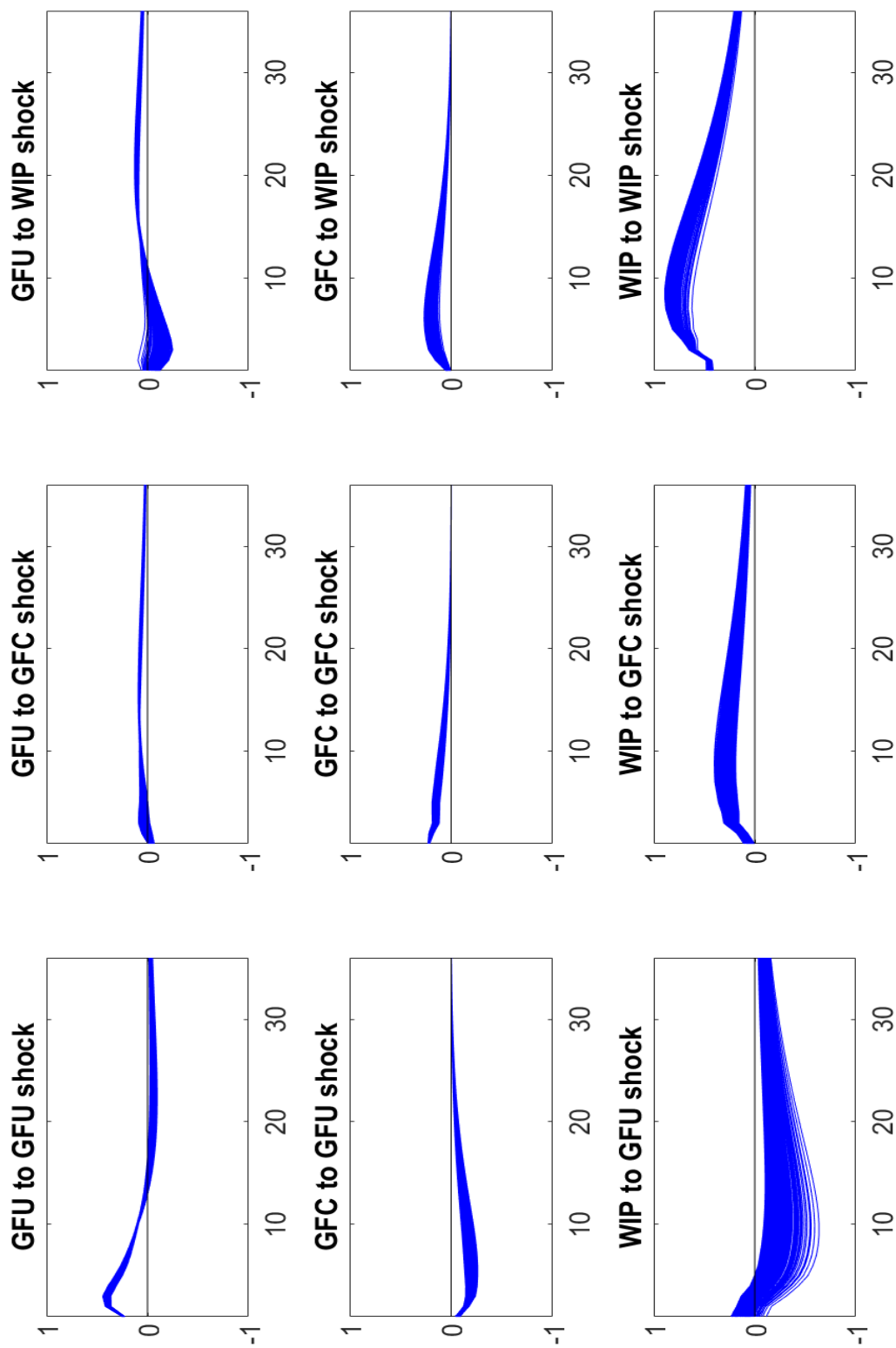


Figure A19: **Impulse Responses to all shocks: Role of ratio restriction related to GFC shocks.** Sample: 1992M7-2019M4. Responses to a one standard deviation shock. Identification of the shocks achieved by imposing the mix of event and sign restrictions indicated in Table 1, with the exception of the ratio restriction involving the GFC shock, which is dropped in this exercise. Measure of global financial cycle estimated as the global component of a dynamic factor model, data: financial returns for 42 countries.